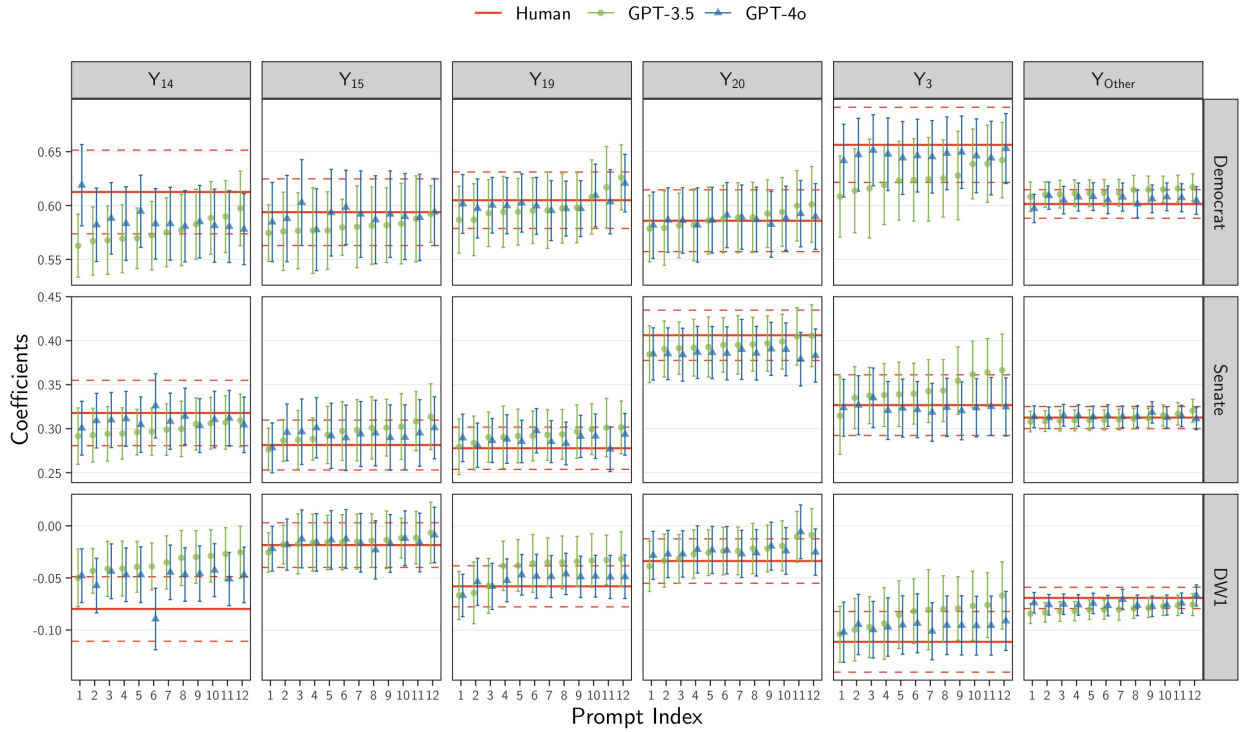


Results RHS

Aug 12, 2024

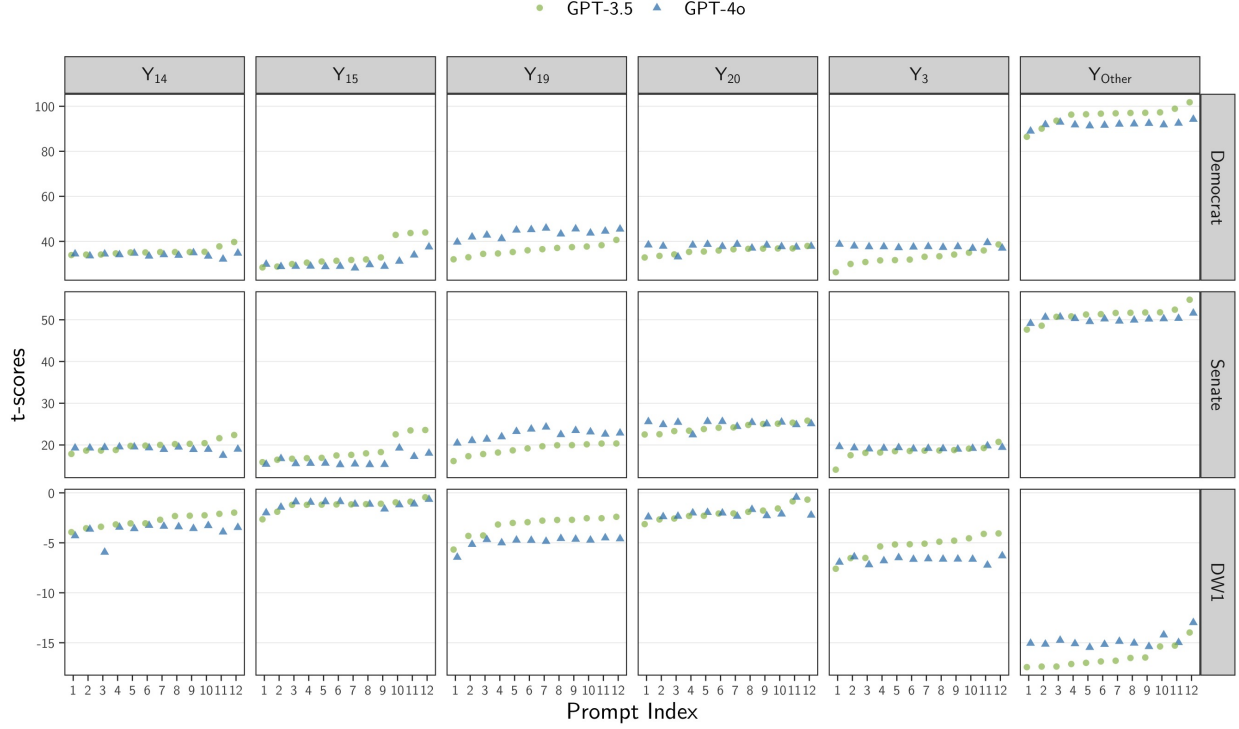
Proxy on the RHS

Figure 1: β_t vs. Prompt for all β s. Proxy on the RHS.



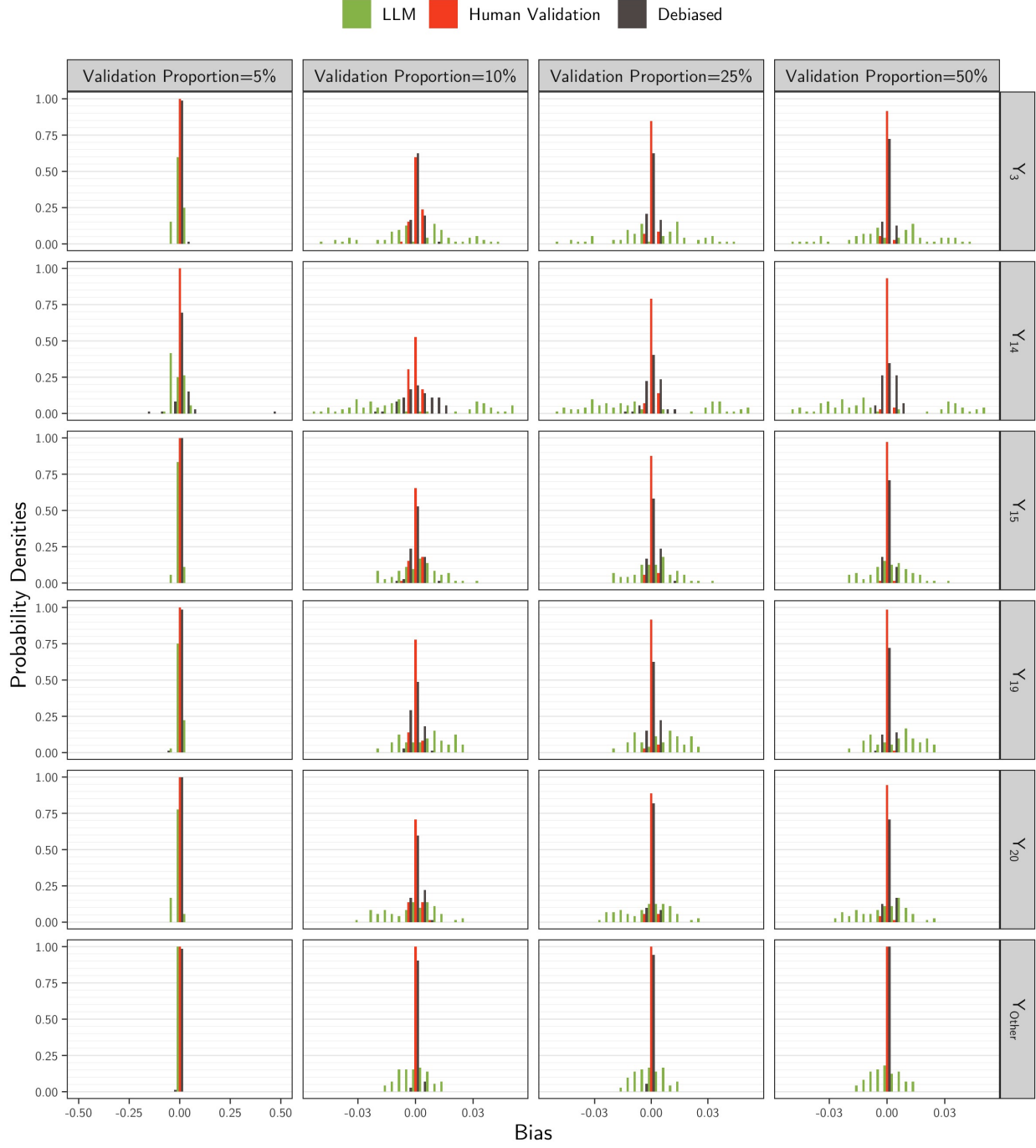
Note: The coefficients were obtained by fitting the model $V = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$ over 10,000 bills, with 95% confidence intervals computed using robust standard errors. Here, Y_t denotes the indicator for topic t , where $Y_t = 1\{Y = t\}$ for $t \in \mathcal{T}$ and $Y_{\text{Other}} = 1\{Y \notin \mathcal{T}\}$, with $\mathcal{T} = \{3, 14, 15, 19, 20\}$ is the 5 most-common human-labeled topics in the bills data. The variable V takes one of the following bills variables: Democrat, Senate, or DW1, where $\text{extDemocrat} = 1\{\text{Party} = \text{Democrat}\}$, $\text{Senate} = 1\{\text{Chamber} = \text{Senate}\}$, and DW1 is the imputed DW1 score for the bill sponsor. The regressions based on human-labeled topics (red) were estimated with $Y_t^{\text{Human}} = 1\{Y^{\text{Human}} = t\}$, while the LLM-based regressions (GPT-3.5 in green, circles; and GPT-4o in blue, triangles) were estimated with $Y_{t,m,p}^{\text{LLM}} = 1\{Y_{m,p}^{\text{LLM}} = t\}$, where $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$ denotes the LLM model and $p \in \{1, \dots, 12\}$ denotes the prompt modifications. In each subplot, the indices for the prompts reflect the order chosen based on sorting the estimated values of the GPT-3.5 model in ascending order. The corresponding estimates for GPT-4o are aligned according to this order.

Figure 2: t-scores of each β_t vs. Prompt. Proxy on the RHS.



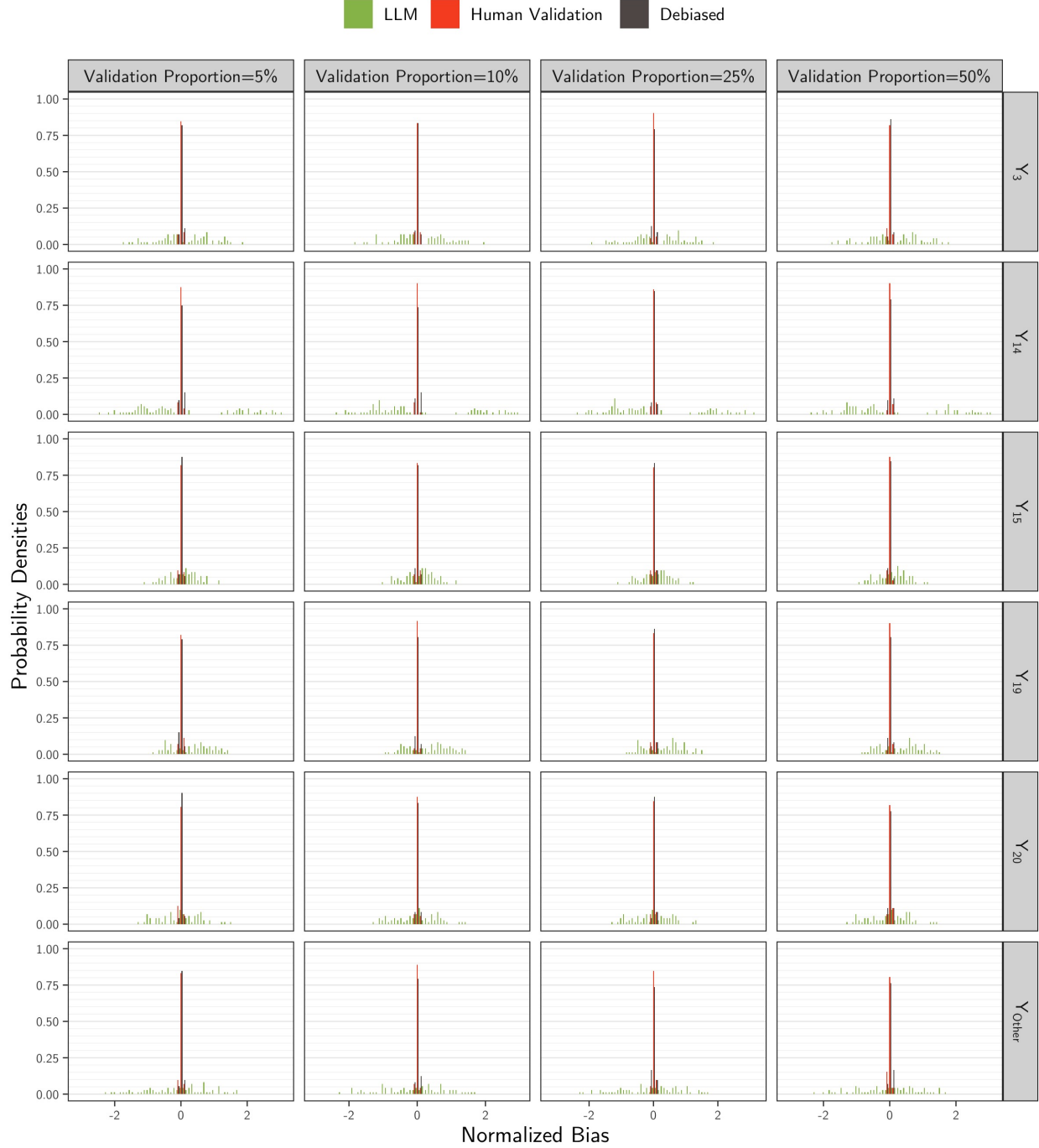
Note: The t-scores were obtained by fitting the model $V = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$ over 10,000 bills, with 95% confidence intervals computed using robust standard errors. Here, Y_t denotes the indicator for topic t , where $Y_t = 1\{Y = t\}$ for $t \in \mathcal{T}$ and $Y_{\text{Other}} = 1\{Y \notin \mathcal{T}\}$, with $\mathcal{T} = \{3, 14, 15, 19, 20\}$ is the 5 most-common human-labeled topics in the bills data. The variable V takes one of the following bills variables: Democrat, Senate, or DW1, where $\text{extDemocrat} = 1\{\text{Party} = \text{Democrat}\}$, $\text{Senate} = 1\{\text{Chamber} = \text{Senate}\}$, and DW1 is the imputed DW1 score for the bill sponsor. The regressions based on human-labeled topics (red) were estimated with $Y_t^{\text{Human}} = 1\{Y^{\text{Human}} = t\}$, while the LLM-based regressions (GPT-3.5 in green, circles; and GPT-4o in blue, triangles) were estimated with $Y_{t,m,p}^{\text{LLM}} = 1\{Y_{m,p}^{\text{LLM}} = t\}$, where $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$ denotes the LLM model and $p \in \{1, \dots, 12\}$ denotes the prompt modifications. In each subplot, the indices for the prompts reflect the order chosen based on sorting the estimated values of the GPT-3.5 model in ascending order. The corresponding estimates for GPT-4o are aligned according to this order.

Figure 3: Bias Density Plots for all β s by Proportion of Validation Samples. Proxy on the RHS.



Note: The bias estimates were computed as the difference between the coefficient estimates from $N = 1000$ simulation runs, $\hat{\beta}_{V,\eta,m,p} = \sum_{i=1}^N \beta_{t,V,\eta,m,p}^{(i)} / N$, and the human-based coefficient from the entire set of 10,000 bills, $\beta_{t,V}^*$, using a regression model of the form $V = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$, for topic $t \in \{3, 14, 15, 19, 20, \text{Other}\}$, variable $V \in \{\text{Democrat, Senate, DW1}\}$, validation proportion $\eta \in \{5\%, 10\%, 25\%, 50\%\}$, LLM model $m \in \{\text{GPT-3.5, GPT-4o}\}$, and prompt $p \in \{1, \dots, 12\}$. For the LLM-based models (green), a random sample of 5,000 bills drawn from the original 10,000 was used, with LLM-based labels, $Y_{t,m,p}^{\text{LLM}} = 1\{Y_{m,p}^{\text{LLM}} = t\}$. For the other two models—human-based (red) and debiased-based (brown)—the 5,000 sample were split into two subsets: validation and primary, with the size of the validation set determined by the validation proportion, η . In the human-based model, only the human labels in the validation set were used, $Y_t^{\text{Human}} = 1\{Y^{\text{Human}} = t\}$, whereas the debiased estimates use both the validation and primary sets, with Y_t^{Human} from validation set used to estimate measurement errors and $Y_{t,m,p}^{\text{LLM}}$ from the primary set used to obtain the coefficient estimates used to compute the reported bias.

Figure 4: Normalized Bias Density of β_1 by Proportion of Validation Samples. Proxy on the LHS.



Note: The normalized bias estimates were computed as the normalized difference between the coefficient estimates from $N = 1000$ simulation runs, $\bar{\beta}_{t,V,\eta,m,p}$, and the human-based coefficient from the entire set of 10,000 bills, $\beta_{t,V}^*$, divided by the sample standard deviation, $s_{t,V,\eta,m,p}$, where $s_{t,V,\eta,m,p}^2 = \sum_{i=1}^N (\beta_{t,V,\eta,m,p}^{(i)} - \bar{\beta}_{t,V,\eta,m,p})^2 / (N - 1)$ and all variables are as defined previously.

Figure 5: Bias Density of β_3 by Model and Proportion of Validation Samples. Proxy on the RHS.



Figure 6: Bias Density of β_{14} by Model and Proportion of Validation Samples. Proxy on the RHS.

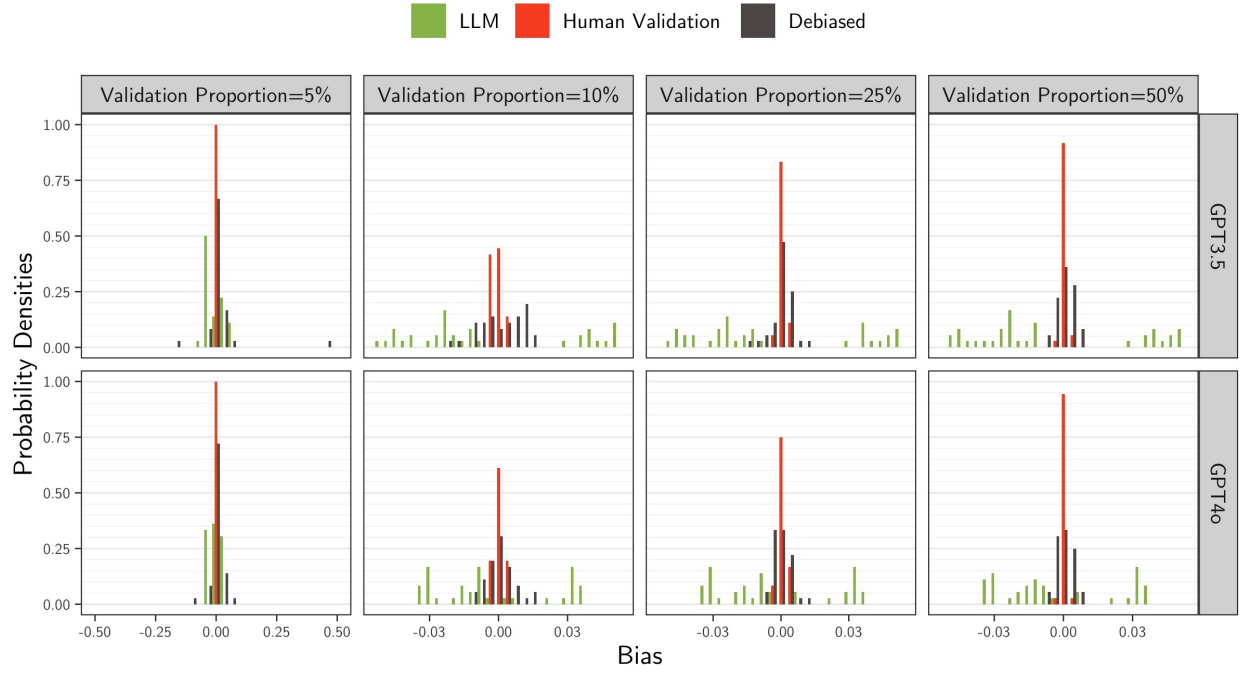


Figure 7: Bias Density of β_{15} by Model and Proportion of Validation Samples. Proxy on the RHS.

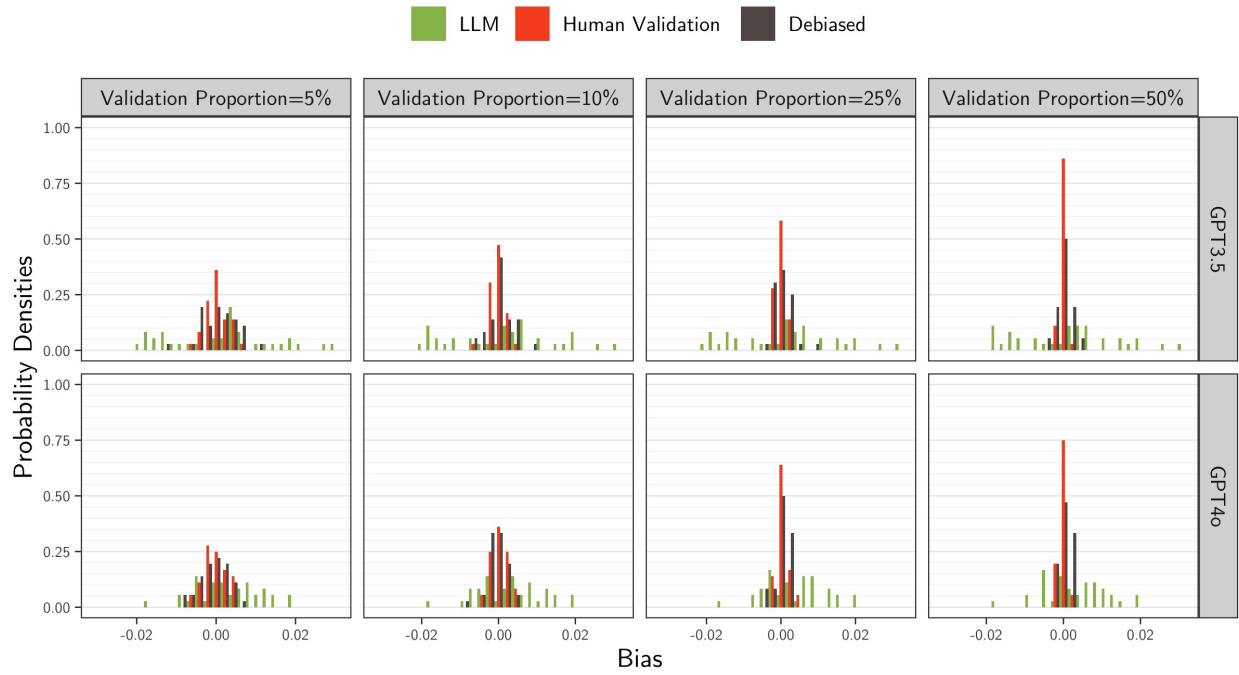


Figure 8: Bias Density of β_{19} by Model and Proportion of Validation Samples. Proxy on the RHS.

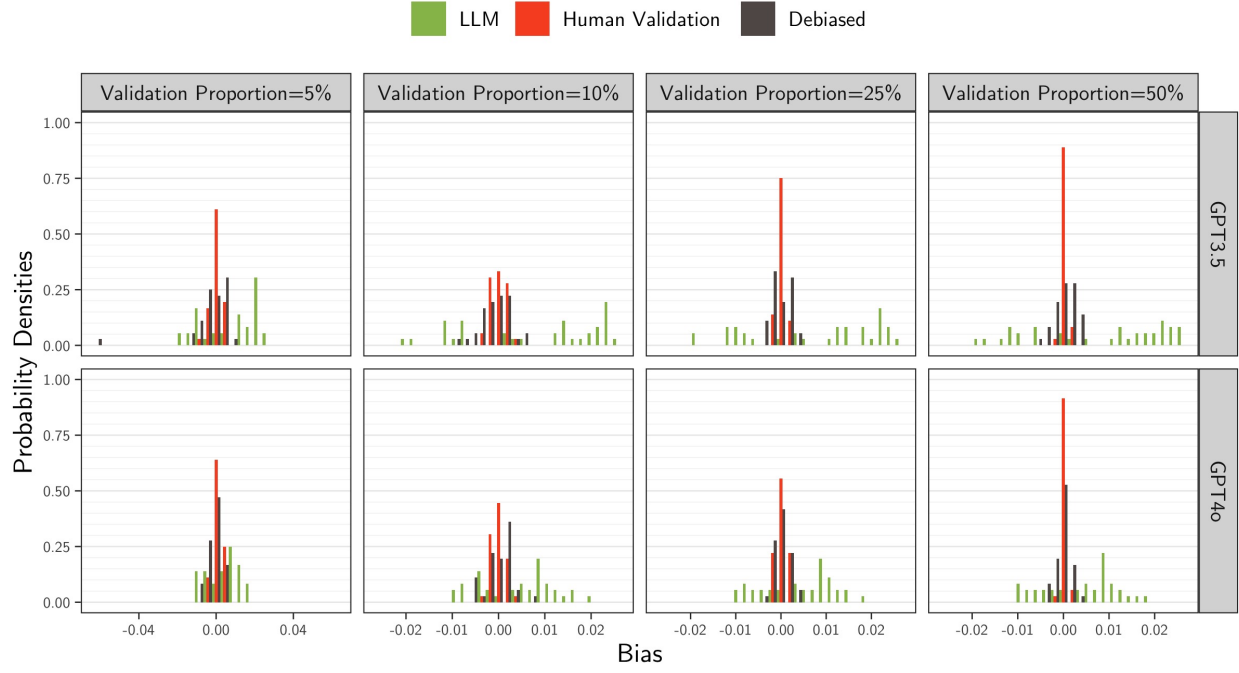


Figure 9: Bias Density of β_{20} by Model and Proportion of Validation Samples. Proxy on the RHS.

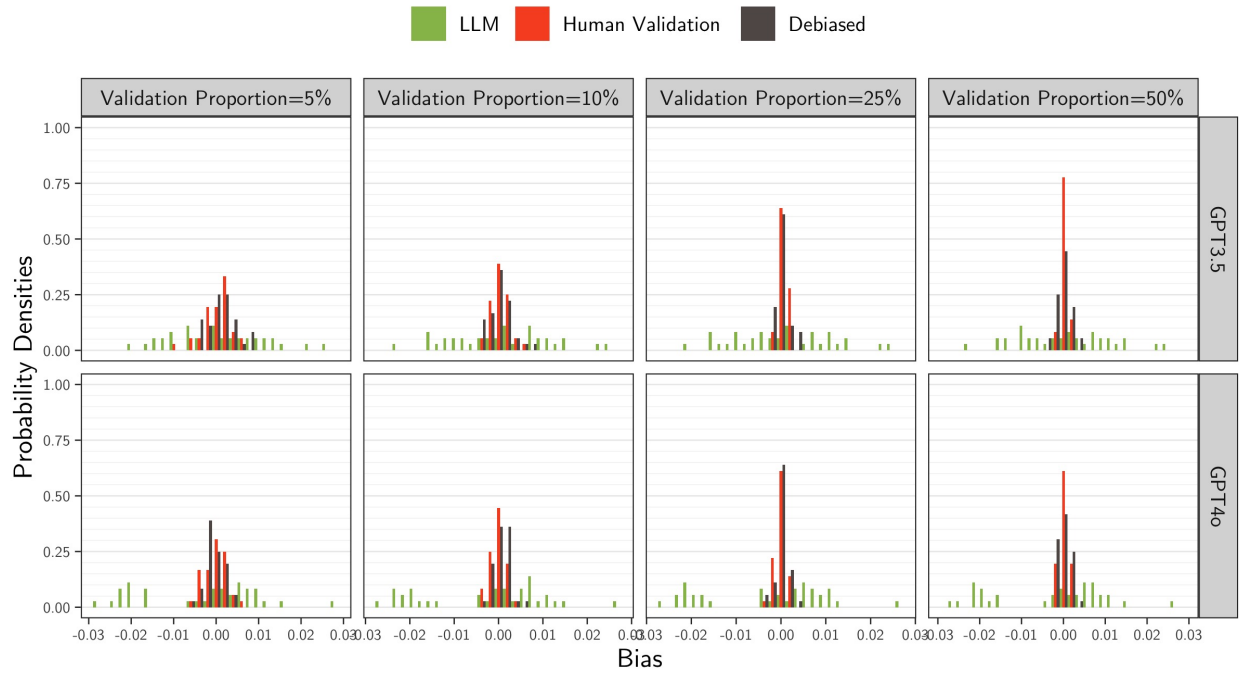


Figure 10: Bias Density of β_{Other} by Model and Proportion of Validation Samples. Proxy on the RHS.

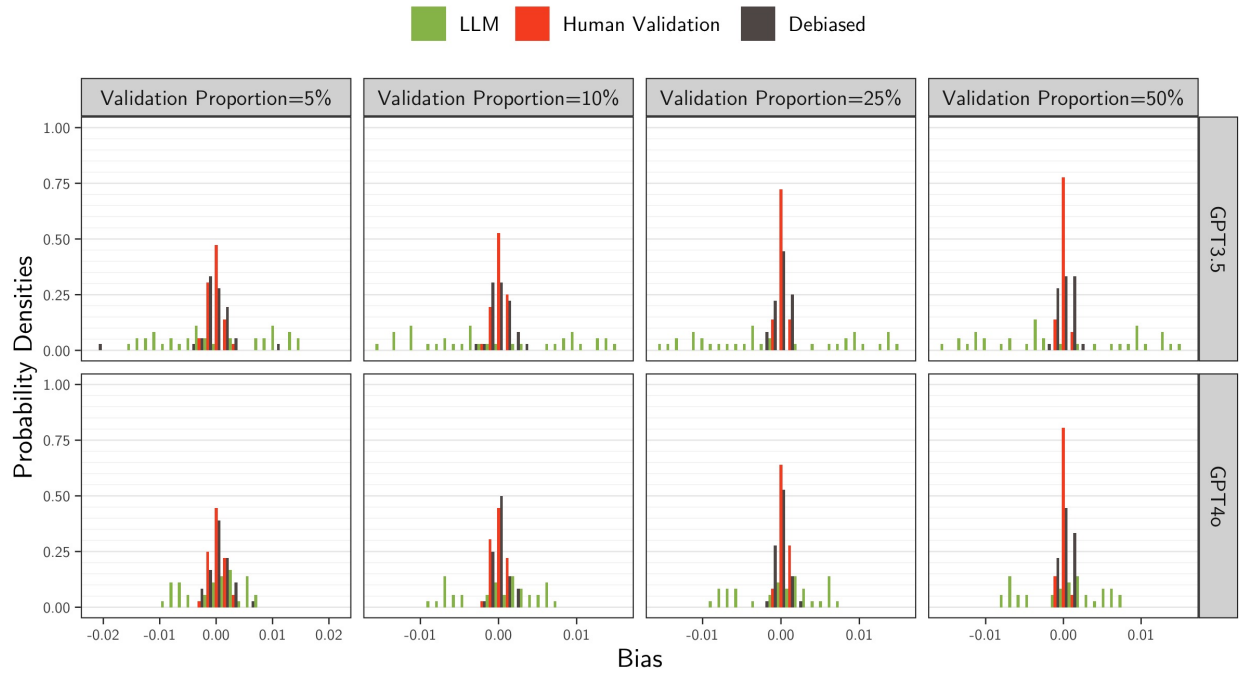


Figure 11: Normalized Bias Density of β_3 by Model and Proportion of Validation Samples. Proxy on the RHS.

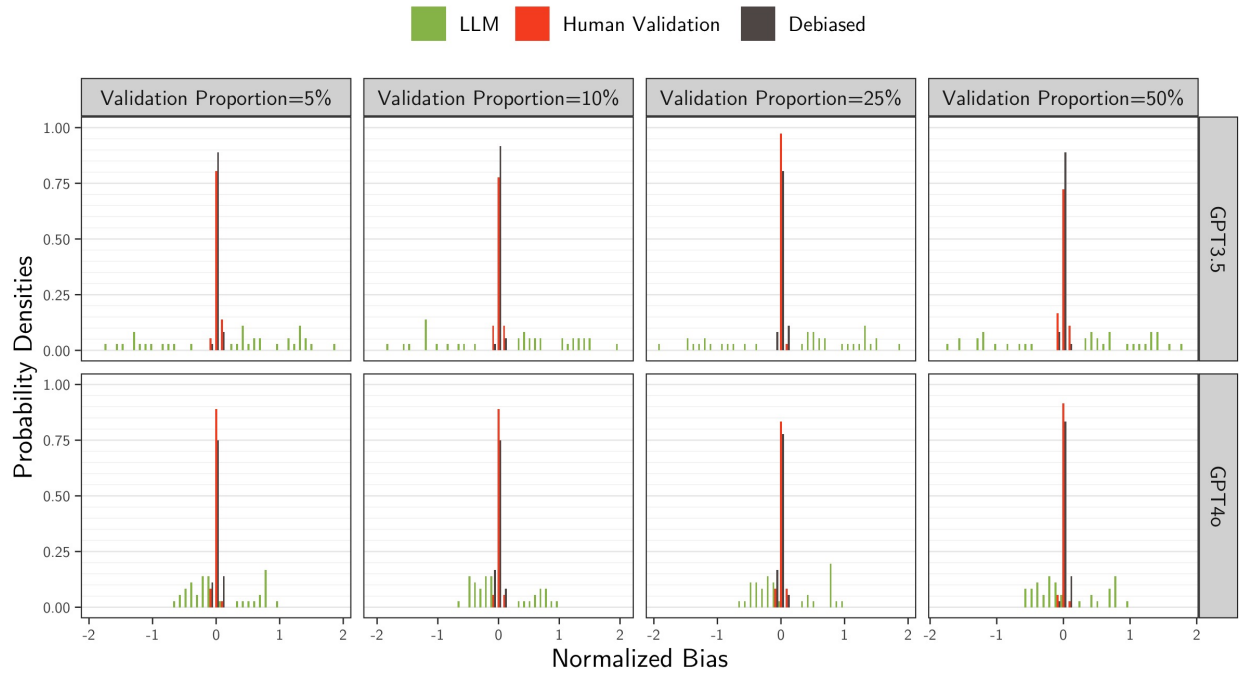


Figure 12: Normalized Bias Density of β_{14} by Model and Proportion of Validation Samples. Proxy on the RHS.

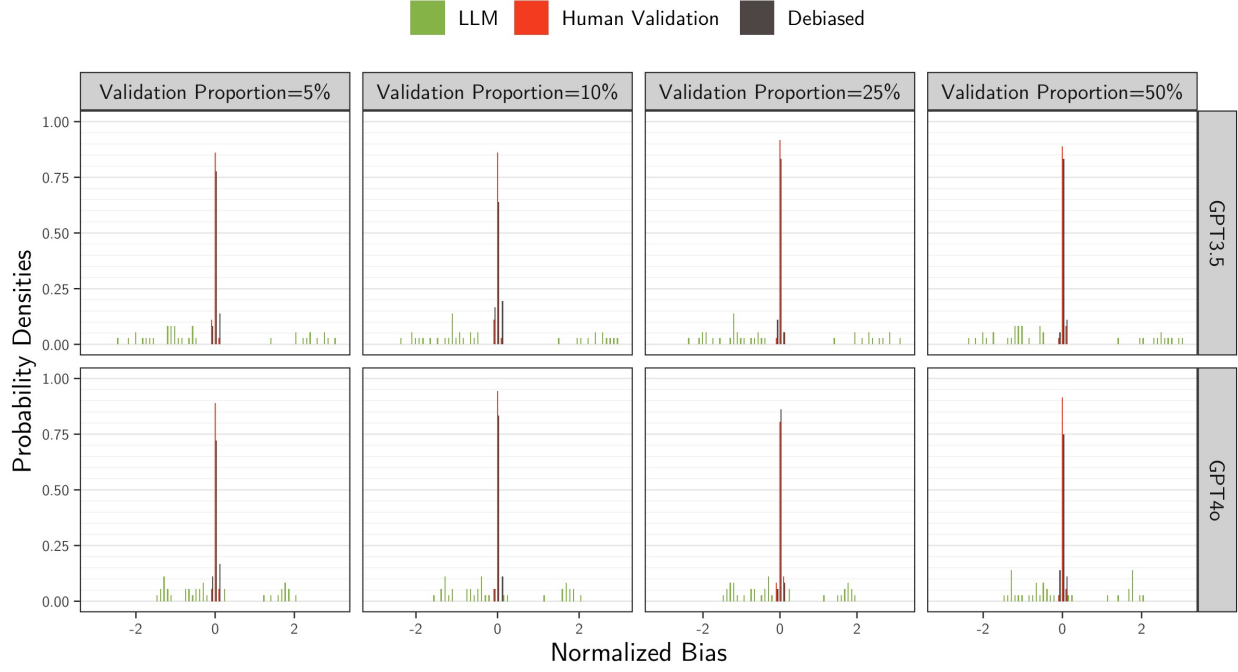


Figure 13: Normalized Bias Density of β_{15} by Model and Proportion of Validation Samples. Proxy on the RHS.

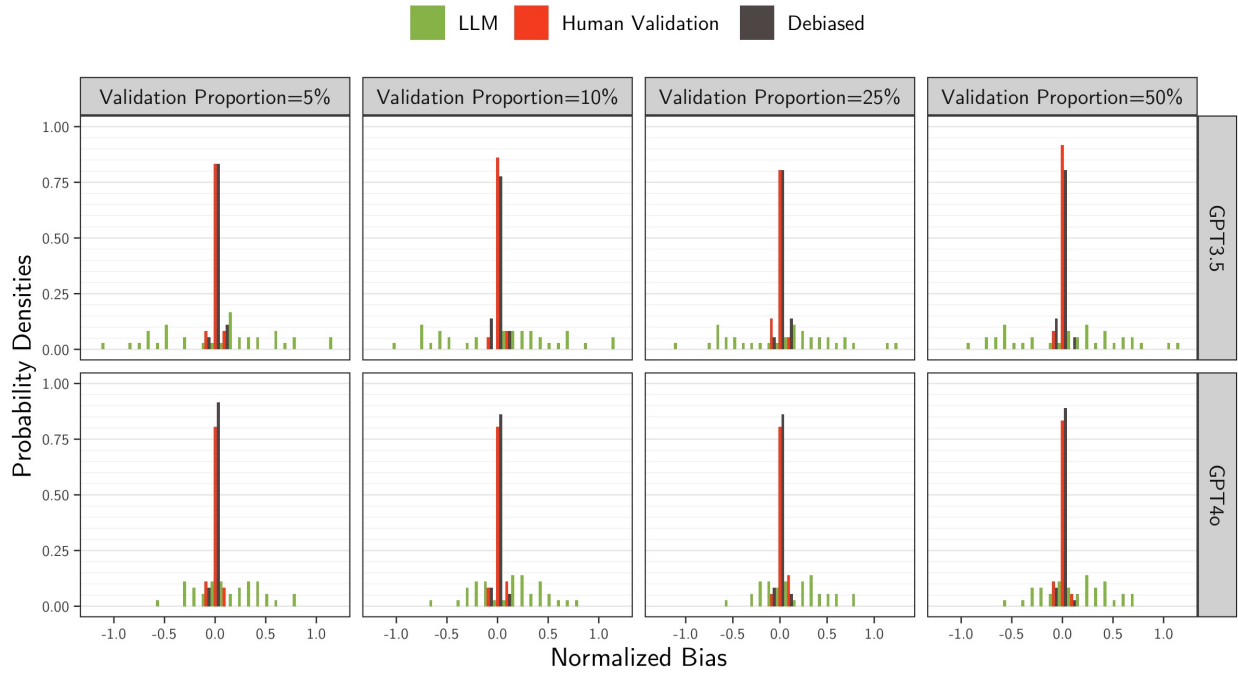


Figure 14: Normalized Bias Density of β_{19} by Model and Proportion of Validation Samples. Proxy on the RHS.



Figure 15: Normalized Bias Density of β_{20} by Model and Proportion of Validation Samples. Proxy on the RHS.

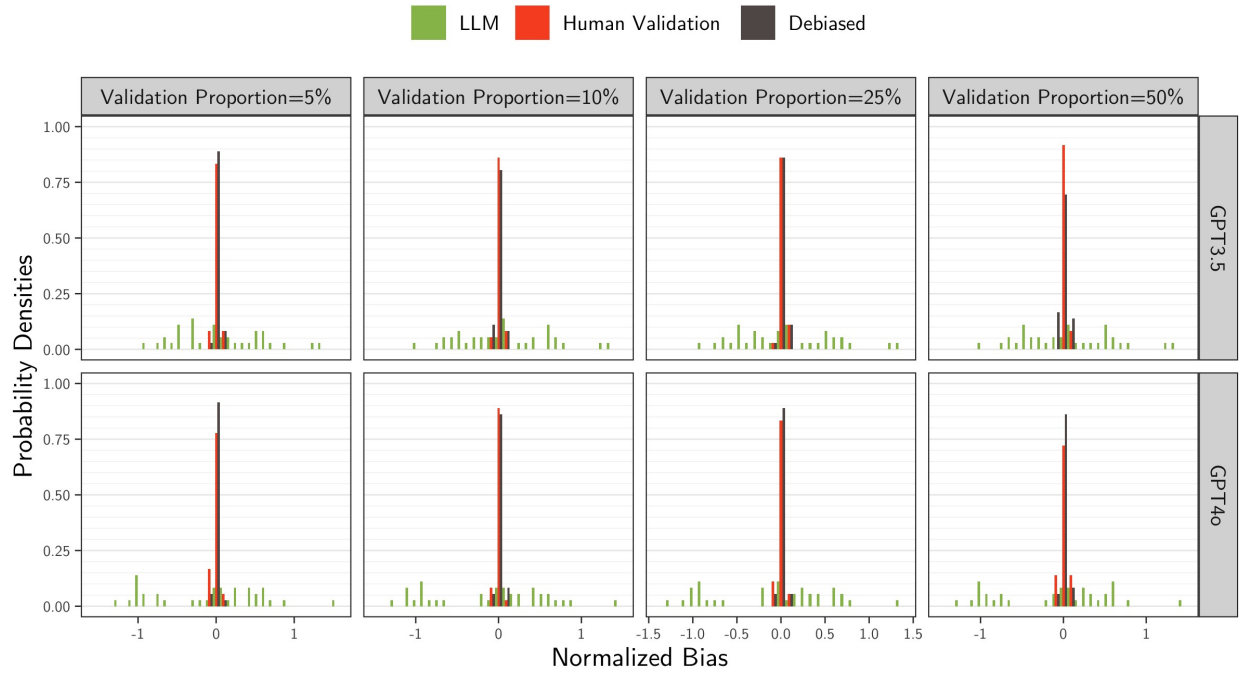


Figure 16: Normalized Bias Density of β_{Other} by Model and Proportion of Validation Samples. Proxy on the RHS.

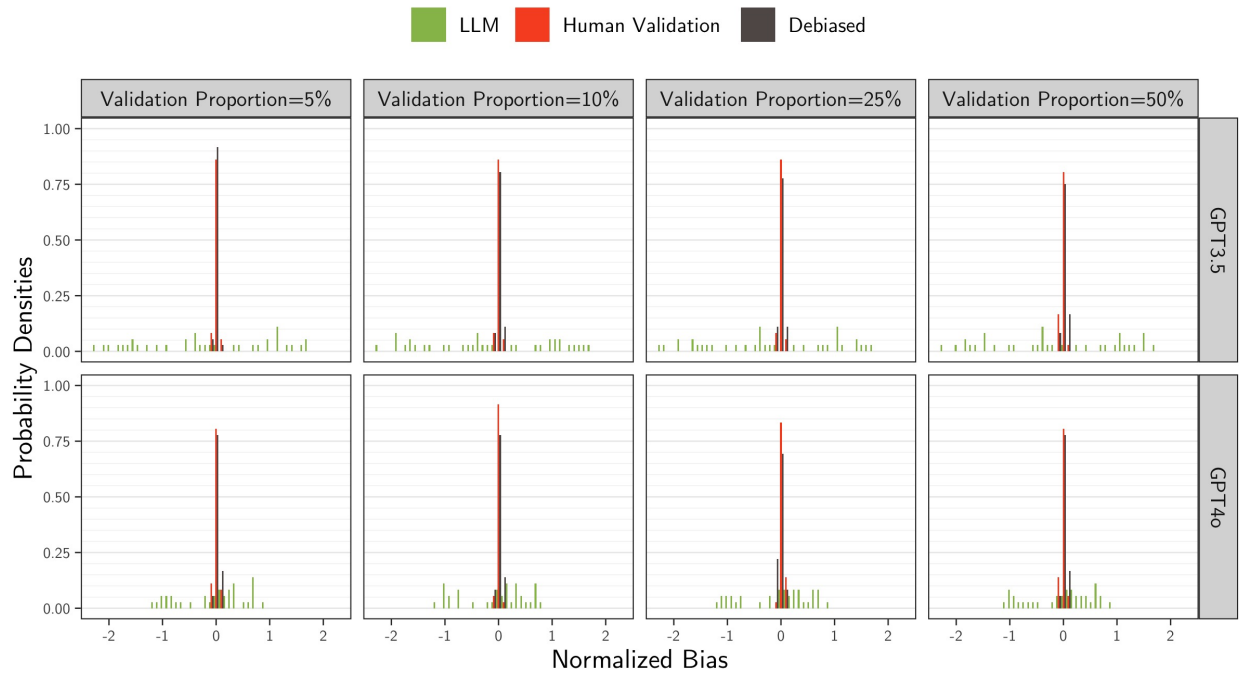


Figure 17: MSE Empirical CDF for all β s by Proportion of Validation Samples. Proxy on the RHS.

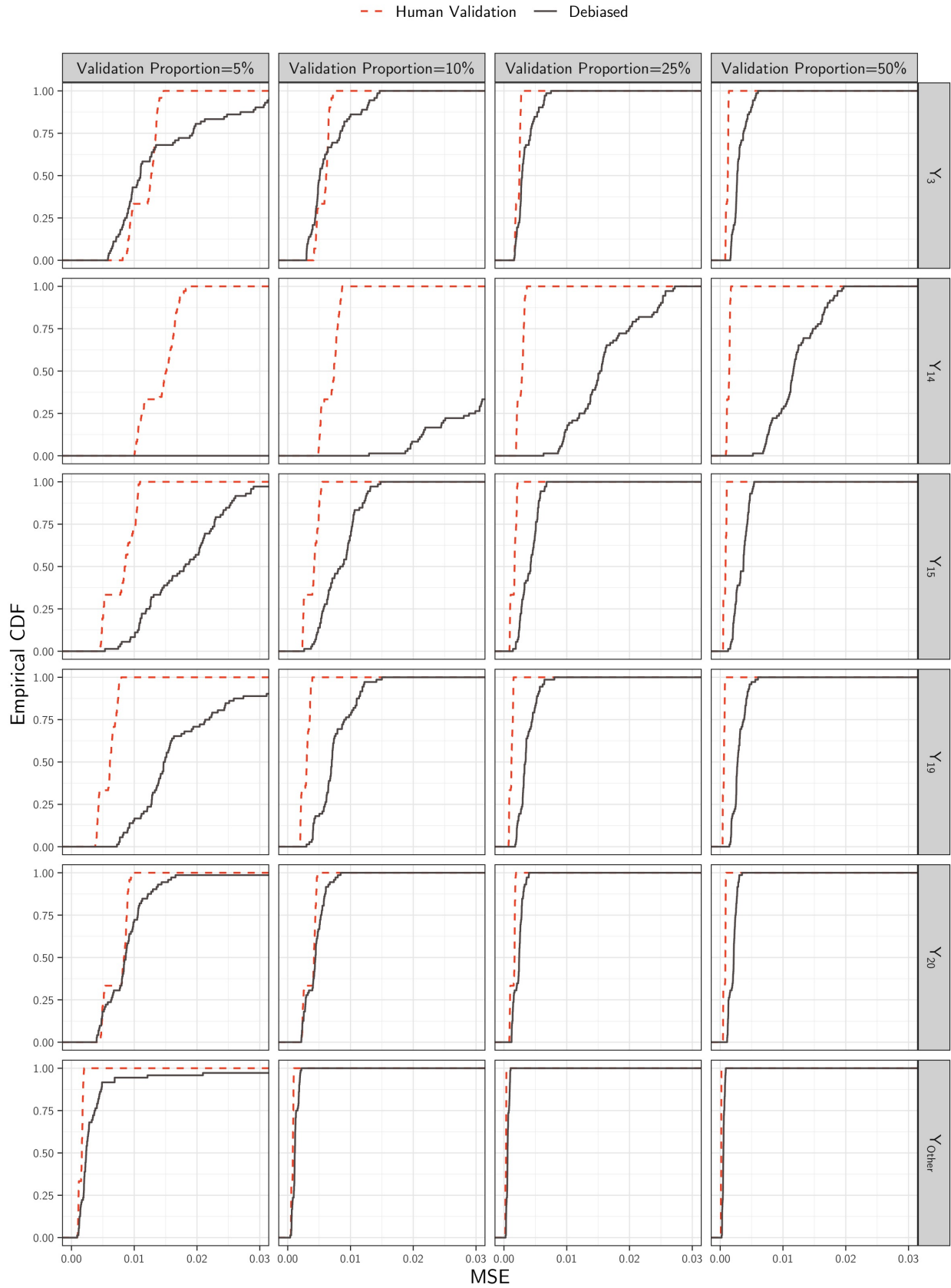


Figure 18: MSE Empirical CDF of β_3 by Model and Proportion of Validation Samples. Proxy on the RHS.

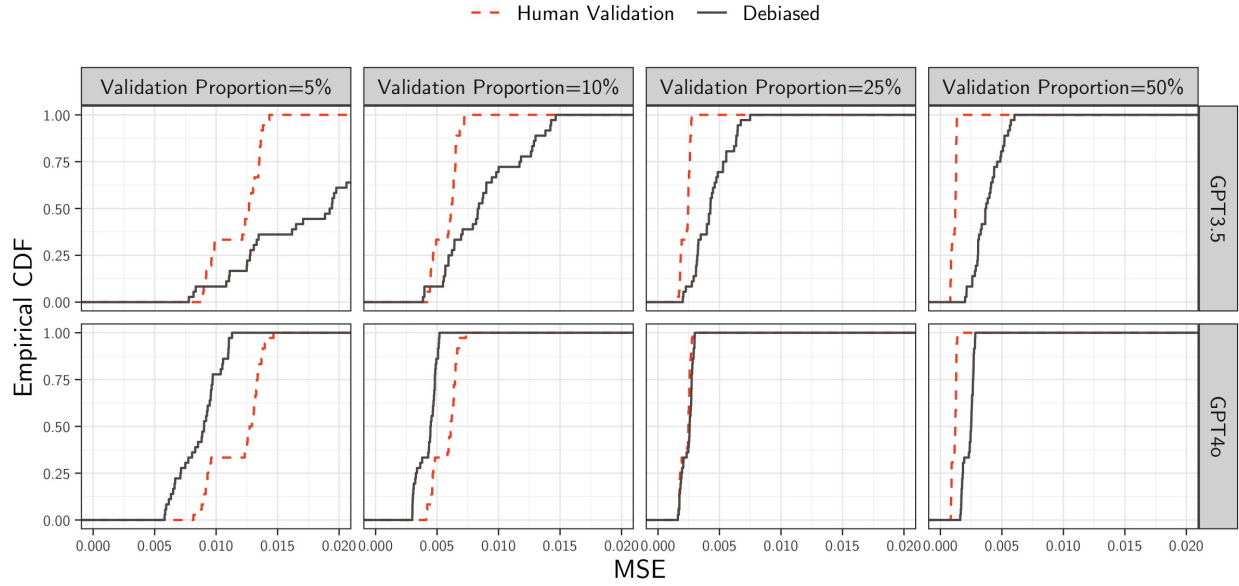


Figure 19: MSE Empirical CDF of β_{14} by Model and Proportion of Validation Samples. Proxy on the RHS.

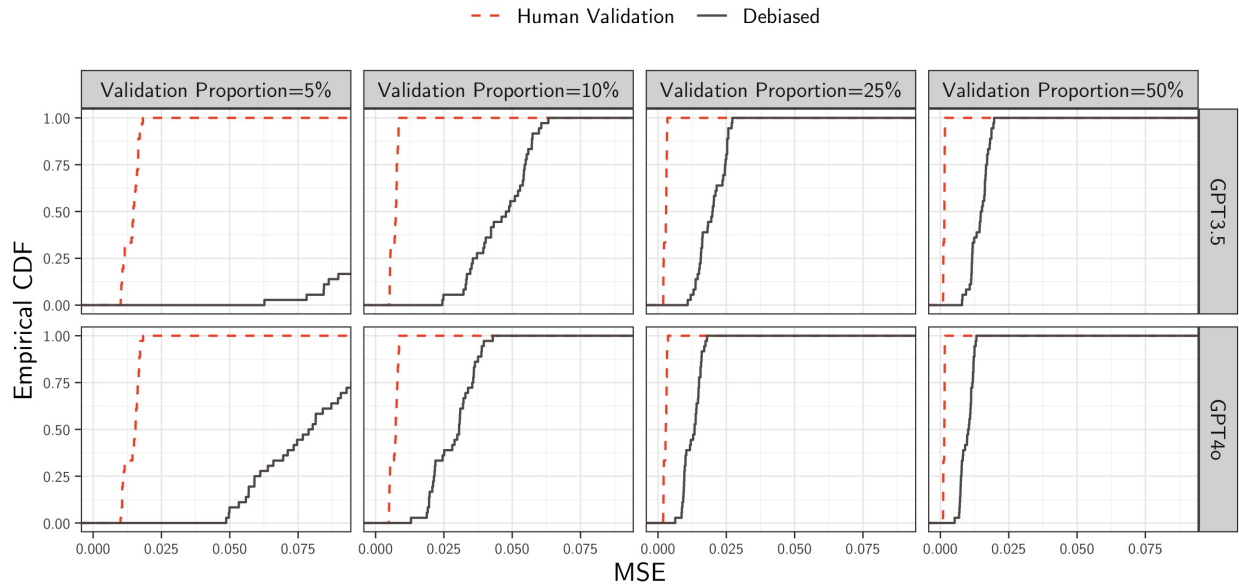


Figure 20: MSE Empirical CDF of β_{15} by Model and Proportion of Validation Samples. Proxy on the RHS.

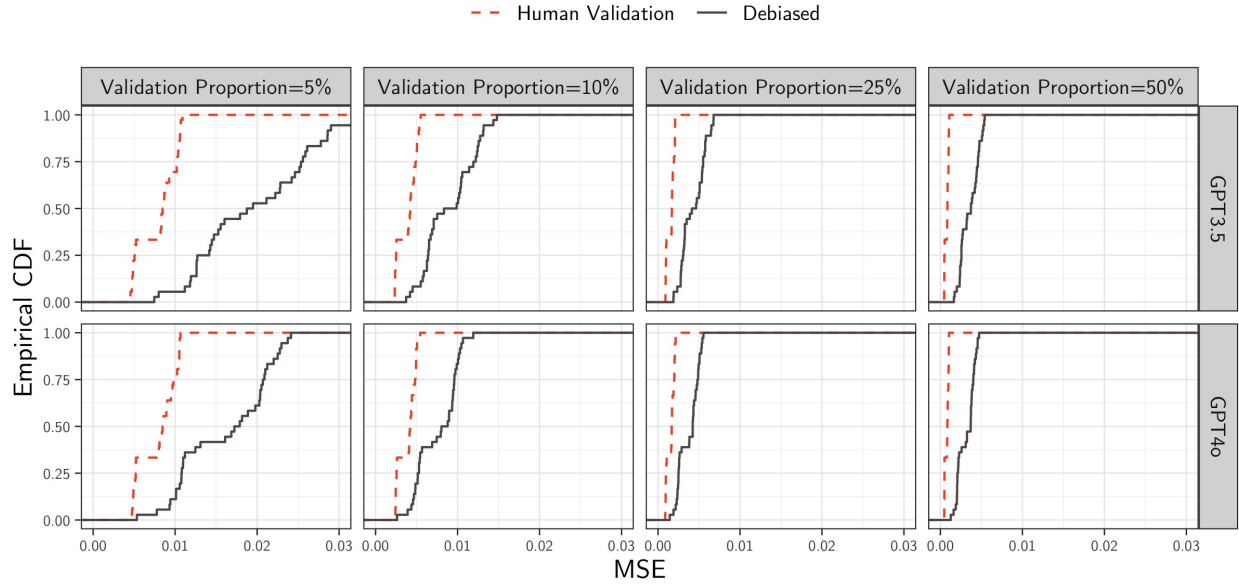


Figure 21: MSE Empirical CDF of β_{19} by Model and Proportion of Validation Samples. Proxy on the RHS.

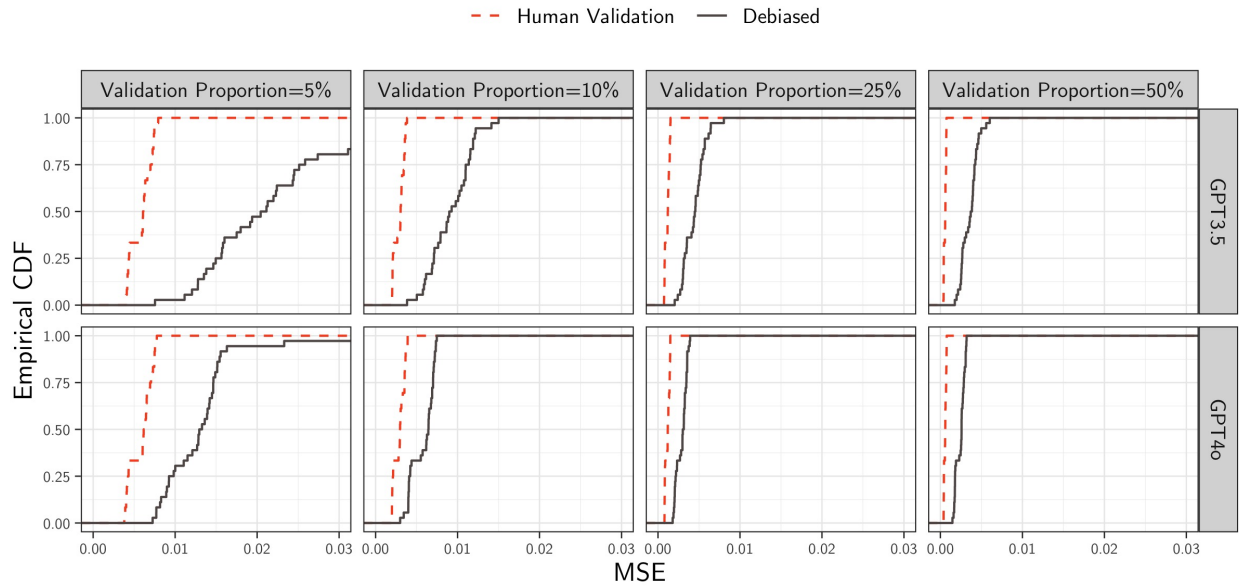


Figure 22: MSE Empirical CDF of β_{20} by Model and Proportion of Validation Samples. Proxy on the RHS.

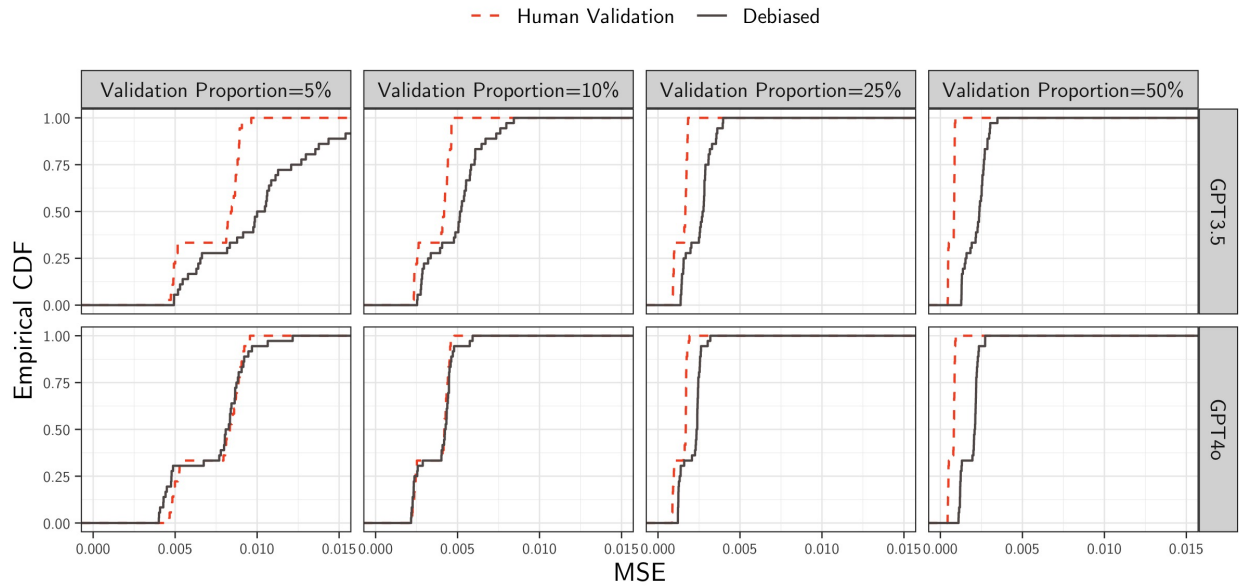


Figure 23: MSE Empirical CDF of β_{Other} by Model and Proportion of Validation Samples. Proxy on the RHS.

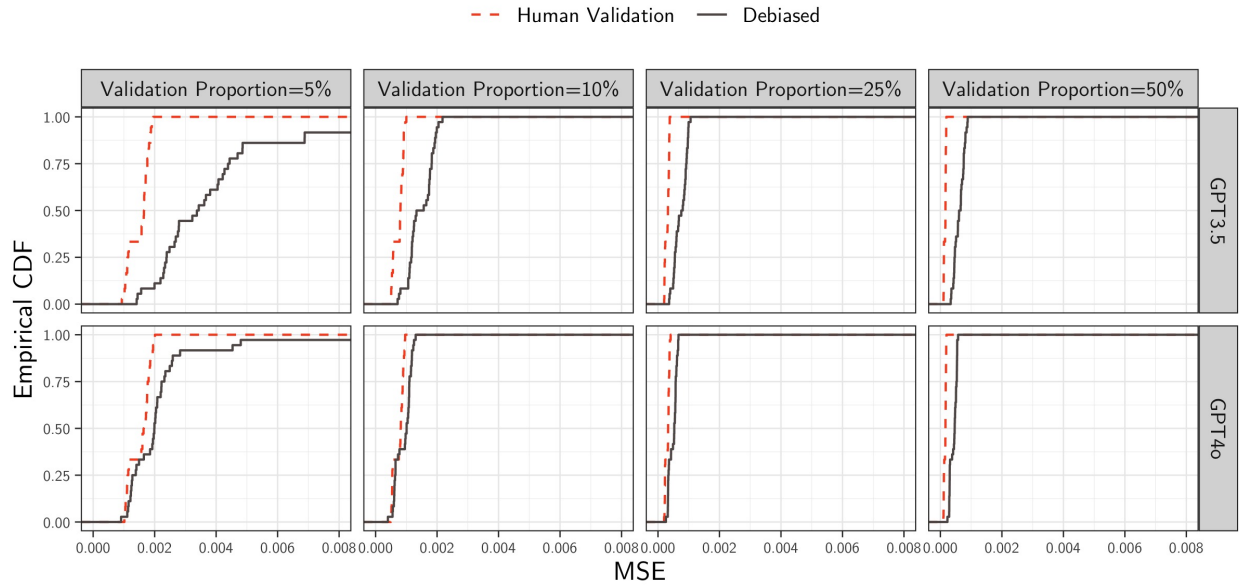


Table 1: Bias Summary Statistics of β_3 by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	0.003	0.021	0.004	-0.035	0.035
	Human Validation	0.000	0.003	0.001	-0.005	0.006
	Debiased	0.001	0.004	0.000	-0.005	0.006
10%	LLM	0.003	0.022	0.003	-0.035	0.035
	Human Validation	0.000	0.002	0.000	-0.004	0.004
	Debiased	0.000	0.003	0.000	-0.004	0.004
25%	LLM	0.003	0.022	0.004	-0.036	0.035
	Human Validation	0.000	0.001	0.000	-0.002	0.002
	Debiased	0.000	0.002	0.000	-0.003	0.003
50%	LLM	0.003	0.022	0.003	-0.036	0.036
	Human Validation	0.000	0.001	0.000	-0.002	0.002
	Debiased	0.000	0.002	0.000	-0.002	0.003

Table 2: Bias Summary Statistics of β_{14} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	-0.003	0.031	-0.012	-0.043	0.049
	Human Validation	0.000	0.003	0.000	-0.006	0.005
	Debiased	0.007	0.063	0.000	-0.030	0.042
10%	LLM	-0.003	0.031	-0.012	-0.043	0.050
	Human Validation	0.000	0.002	-0.001	-0.004	0.003
	Debiased	0.001	0.008	0.001	-0.010	0.013
25%	LLM	-0.003	0.031	-0.012	-0.043	0.050
	Human Validation	0.000	0.002	0.000	-0.003	0.003
	Debiased	0.000	0.004	0.001	-0.007	0.006
50%	LLM	-0.003	0.031	-0.012	-0.043	0.048
	Human Validation	0.000	0.001	0.000	-0.001	0.002
	Debiased	0.000	0.004	0.000	-0.005	0.006

Table 3: Bias Summary Statistics of β_{15} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	0.003	0.011	0.003	-0.017	0.020
	Human Validation	0.000	0.003	0.000	-0.005	0.004
	Debiased	0.000	0.004	0.000	-0.007	0.006
10%	LLM	0.002	0.011	0.003	-0.018	0.020
	Human Validation	0.000	0.002	0.000	-0.003	0.003
	Debiased	0.000	0.003	0.000	-0.005	0.004
25%	LLM	0.003	0.011	0.003	-0.017	0.020
	Human Validation	0.000	0.001	0.000	-0.002	0.002
	Debiased	0.000	0.002	0.001	-0.003	0.003
50%	LLM	0.003	0.011	0.003	-0.017	0.019
	Human Validation	0.000	0.001	0.000	-0.002	0.001
	Debiased	0.000	0.002	0.000	-0.003	0.003

Table 4: Bias Summary Statistics of β_{19} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	0.007	0.012	0.009	-0.011	0.023
	Human Validation	0.001	0.003	0.001	-0.004	0.005
	Debiased	-0.002	0.009	-0.001	-0.010	0.005
10%	LLM	0.007	0.012	0.009	-0.011	0.024
	Human Validation	0.000	0.002	0.000	-0.003	0.002
	Debiased	0.000	0.003	0.000	-0.005	0.004
25%	LLM	0.006	0.012	0.009	-0.011	0.024
	Human Validation	0.000	0.001	0.000	-0.002	0.002
	Debiased	0.000	0.002	0.000	-0.003	0.003
50%	LLM	0.006	0.012	0.009	-0.011	0.024
	Human Validation	0.000	0.001	0.000	-0.001	0.001
	Debiased	0.000	0.002	0.000	-0.003	0.003

Table 5: Bias Summary Statistics of β_{20} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	-0.001	0.013	0.001	-0.022	0.016
	Human Validation	0.000	0.003	0.000	-0.005	0.004
	Debiased	0.000	0.003	0.000	-0.004	0.005
10%	LLM	-0.001	0.013	0.001	-0.023	0.015
	Human Validation	0.000	0.002	0.000	-0.003	0.003
	Debiased	0.000	0.002	0.000	-0.003	0.004
25%	LLM	-0.001	0.013	0.001	-0.021	0.015
	Human Validation	0.000	0.001	0.000	-0.002	0.002
	Debiased	0.000	0.001	0.000	-0.002	0.003
50%	LLM	-0.001	0.012	0.001	-0.021	0.015
	Human Validation	0.000	0.001	0.000	-0.001	0.002
	Debiased	0.000	0.002	0.000	-0.003	0.003

Table 6: Bias Summary Statistics of β_{Other} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	0	0.008	0	-0.012	0.013
	Human Validation	0	0.001	0	-0.002	0.002
	Debiased	0	0.003	0	-0.003	0.003
10%	LLM	0	0.008	0	-0.012	0.013
	Human Validation	0	0.001	0	-0.001	0.001
	Debiased	0	0.001	0	-0.002	0.002
25%	LLM	0	0.008	0	-0.012	0.014
	Human Validation	0	0.001	0	-0.001	0.001
	Debiased	0	0.001	0	-0.002	0.001
50%	LLM	0	0.008	0	-0.012	0.013
	Human Validation	0	0.000	0	-0.001	0.001
	Debiased	0	0.001	0	-0.001	0.001

Table 7: Normalized Bias Summary Statistics of β_3 by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	0.148	0.846	0.166	-1.236	1.398
5%	Human Validation	0.001	0.032	0.005	-0.052	0.057
5%	Debiased	0.003	0.033	0.003	-0.051	0.051
10%	LLM	0.150	0.842	0.139	-1.202	1.419
10%	Human Validation	-0.002	0.029	-0.006	-0.051	0.049
10%	Debiased	0.000	0.033	-0.003	-0.056	0.050
25%	LLM	0.154	0.852	0.165	-1.287	1.419
25%	Human Validation	0.006	0.028	0.008	-0.040	0.047
25%	Debiased	-0.001	0.035	-0.001	-0.055	0.057
50%	LLM	0.150	0.845	0.124	-1.286	1.444
50%	Human Validation	0.001	0.032	0.003	-0.053	0.045
50%	Debiased	-0.002	0.029	-0.004	-0.046	0.047

Table 8: Normalized Bias Summary Statistics of β_{14} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	0.013	1.541	-0.537	-1.885	2.673
5%	Human Validation	-0.002	0.029	-0.001	-0.050	0.044
5%	Debiased	0.004	0.039	0.002	-0.057	0.063
10%	LLM	0.013	1.543	-0.553	-1.929	2.667
10%	Human Validation	-0.006	0.028	-0.006	-0.046	0.043
10%	Debiased	0.002	0.039	0.003	-0.058	0.067
25%	LLM	0.005	1.541	-0.560	-1.934	2.622
25%	Human Validation	0.001	0.032	-0.001	-0.047	0.050
25%	Debiased	-0.001	0.034	0.005	-0.059	0.056
50%	LLM	0.009	1.547	-0.543	-1.968	2.667
50%	Human Validation	0.003	0.032	0.004	-0.042	0.050
50%	Debiased	0.001	0.034	0.001	-0.057	0.057

Table 9: Normalized Bias Summary Statistics of β_{15} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	0.118	0.463	0.166	-0.633	0.791
5%	Human Validation	-0.003	0.033	-0.004	-0.056	0.050
5%	Debiased	-0.002	0.030	-0.002	-0.046	0.045
10%	LLM	0.108	0.470	0.167	-0.695	0.764
10%	Human Validation	-0.001	0.034	0.000	-0.047	0.056
10%	Debiased	-0.003	0.033	-0.002	-0.060	0.048
25%	LLM	0.121	0.460	0.150	-0.653	0.782
25%	Human Validation	-0.005	0.034	-0.012	-0.055	0.056
25%	Debiased	0.004	0.035	0.007	-0.052	0.054
50%	LLM	0.112	0.450	0.158	-0.649	0.762
50%	Human Validation	-0.006	0.028	-0.003	-0.052	0.035
50%	Debiased	-0.002	0.030	0.000	-0.051	0.042

Table 10: Normalized Bias Summary Statistics of β_{19} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	0.369	0.615	0.486	-0.534	1.285
	Human Validation	0.008	0.034	0.010	-0.050	0.062
	Debiased	-0.010	0.034	-0.009	-0.064	0.039
10%	LLM	0.368	0.615	0.466	-0.502	1.344
	Human Validation	-0.004	0.029	-0.006	-0.044	0.039
	Debiased	-0.004	0.034	-0.003	-0.064	0.048
25%	LLM	0.359	0.607	0.451	-0.493	1.326
	Human Validation	-0.001	0.032	-0.001	-0.058	0.055
	Debiased	0.000	0.033	-0.002	-0.045	0.054
50%	LLM	0.361	0.619	0.466	-0.554	1.331
	Human Validation	0.007	0.027	0.007	-0.034	0.047
	Debiased	-0.002	0.035	0.000	-0.069	0.049

Table 11: Normalized Bias Summary Statistics of β_{20} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	-0.008	0.632	0.023	-1.000	0.888
	Human Validation	-0.001	0.034	0.001	-0.059	0.047
	Debiased	0.002	0.030	0.001	-0.041	0.048
10%	LLM	-0.002	0.630	0.041	-1.005	0.864
	Human Validation	0.001	0.030	0.002	-0.049	0.043
	Debiased	0.004	0.033	0.004	-0.048	0.056
25%	LLM	-0.009	0.623	0.029	-0.970	0.838
	Human Validation	0.003	0.030	0.004	-0.049	0.047
	Debiased	-0.001	0.030	-0.007	-0.040	0.060
50%	LLM	-0.003	0.619	0.049	-0.990	0.835
	Human Validation	0.001	0.034	0.000	-0.055	0.054
	Debiased	0.000	0.036	0.003	-0.055	0.061

Table 12: Normalized Bias Summary Statistics of β_{Other} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	-0.093	0.979	0.030	-1.766	1.413
	Human Validation	-0.003	0.033	-0.006	-0.051	0.055
	Debiased	0.001	0.031	-0.006	-0.046	0.055
10%	LLM	-0.100	0.961	0.021	-1.788	1.394
	Human Validation	-0.001	0.030	0.001	-0.049	0.044
	Debiased	0.000	0.035	-0.004	-0.049	0.060
25%	LLM	-0.100	0.970	0.016	-1.767	1.439
	Human Validation	0.001	0.030	-0.002	-0.045	0.053
	Debiased	-0.005	0.036	-0.004	-0.065	0.055
50%	LLM	-0.082	0.957	0.036	-1.721	1.424
	Human Validation	-0.001	0.034	0.008	-0.057	0.043
	Debiased	0.004	0.037	0.001	-0.050	0.060

Table 13: Bias Summary Statistics of β_3 by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
NA	5%	LLM	0.003	0.021	0.004	-0.035	0.035
NA	5%	Human Validation	0.000	0.003	0.001	-0.005	0.006
NA	5%	Debiased	0.001	0.004	0.000	-0.005	0.006
NA	10%	LLM	0.003	0.022	0.003	-0.035	0.035
NA	10%	Human Validation	0.000	0.002	0.000	-0.004	0.004
NA	10%	Debiased	0.000	0.003	0.000	-0.004	0.004
NA	25%	LLM	0.003	0.022	0.004	-0.036	0.035
NA	25%	Human Validation	0.000	0.001	0.000	-0.002	0.002
NA	25%	Debiased	0.000	0.002	0.000	-0.003	0.003
NA	50%	LLM	0.003	0.022	0.003	-0.036	0.036
NA	50%	Human Validation	0.000	0.001	0.000	-0.002	0.002
NA	50%	Debiased	0.000	0.002	0.000	-0.002	0.003

Table 14: Bias Summary Statistics of β_{14} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
NA	5%	LLM	-0.003	0.031	-0.012	-0.043	0.049
NA	5%	Human Validation	0.000	0.003	0.000	-0.006	0.005
NA	5%	Debiased	0.007	0.063	0.000	-0.030	0.042
NA	10%	LLM	-0.003	0.031	-0.012	-0.043	0.050
NA	10%	Human Validation	0.000	0.002	-0.001	-0.004	0.003
NA	10%	Debiased	0.001	0.008	0.001	-0.010	0.013
NA	25%	LLM	-0.003	0.031	-0.012	-0.043	0.050
NA	25%	Human Validation	0.000	0.002	0.000	-0.003	0.003
NA	25%	Debiased	0.000	0.004	0.001	-0.007	0.006
NA	50%	LLM	-0.003	0.031	-0.012	-0.043	0.048
NA	50%	Human Validation	0.000	0.001	0.000	-0.001	0.002
NA	50%	Debiased	0.000	0.004	0.000	-0.005	0.006

Table 15: Bias Summary Statistics of β_{15} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
NA	5%	LLM	0.003	0.011	0.003	-0.017	0.020
NA	5%	Human Validation	0.000	0.003	0.000	-0.005	0.004
NA	5%	Debiased	0.000	0.004	0.000	-0.007	0.006
NA	10%	LLM	0.002	0.011	0.003	-0.018	0.020
NA	10%	Human Validation	0.000	0.002	0.000	-0.003	0.003
NA	10%	Debiased	0.000	0.003	0.000	-0.005	0.004
NA	25%	LLM	0.003	0.011	0.003	-0.017	0.020
NA	25%	Human Validation	0.000	0.001	0.000	-0.002	0.002
NA	25%	Debiased	0.000	0.002	0.001	-0.003	0.003
NA	50%	LLM	0.003	0.011	0.003	-0.017	0.019
NA	50%	Human Validation	0.000	0.001	0.000	-0.002	0.001
NA	50%	Debiased	0.000	0.002	0.000	-0.003	0.003

Table 16: Bias Summary Statistics of β_{19} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
NA	5%	LLM	0.007	0.012	0.009	-0.011	0.023
NA	5%	Human Validation	0.001	0.003	0.001	-0.004	0.005
NA	5%	Debiased	-0.002	0.009	-0.001	-0.010	0.005
NA	10%	LLM	0.007	0.012	0.009	-0.011	0.024
NA	10%	Human Validation	0.000	0.002	0.000	-0.003	0.002
NA	10%	Debiased	0.000	0.003	0.000	-0.005	0.004
NA	25%	LLM	0.006	0.012	0.009	-0.011	0.024
NA	25%	Human Validation	0.000	0.001	0.000	-0.002	0.002
NA	25%	Debiased	0.000	0.002	0.000	-0.003	0.003
NA	50%	LLM	0.006	0.012	0.009	-0.011	0.024
NA	50%	Human Validation	0.000	0.001	0.000	-0.001	0.001
NA	50%	Debiased	0.000	0.002	0.000	-0.003	0.003

Table 17: Bias Summary Statistics of β_{20} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
NA	5%	LLM	-0.001	0.013	0.001	-0.022	0.016
NA	5%	Human Validation	0.000	0.003	0.000	-0.005	0.004
NA	5%	Debiased	0.000	0.003	0.000	-0.004	0.005
NA	10%	LLM	-0.001	0.013	0.001	-0.023	0.015
NA	10%	Human Validation	0.000	0.002	0.000	-0.003	0.003
NA	10%	Debiased	0.000	0.002	0.000	-0.003	0.004
NA	25%	LLM	-0.001	0.013	0.001	-0.021	0.015
NA	25%	Human Validation	0.000	0.001	0.000	-0.002	0.002
NA	25%	Debiased	0.000	0.001	0.000	-0.002	0.003
NA	50%	LLM	-0.001	0.012	0.001	-0.021	0.015
NA	50%	Human Validation	0.000	0.001	0.000	-0.001	0.002
NA	50%	Debiased	0.000	0.002	0.000	-0.003	0.003

Table 18: Bias Summary Statistics of β_{Other} by Model and Proportion of Validation Samples for. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
NA	5%	LLM	0	0.008	0	-0.012	0.013
NA	5%	Human Validation	0	0.001	0	-0.002	0.002
NA	5%	Debiased	0	0.003	0	-0.003	0.003
NA	10%	LLM	0	0.008	0	-0.012	0.013
NA	10%	Human Validation	0	0.001	0	-0.001	0.001
NA	10%	Debiased	0	0.001	0	-0.002	0.002
NA	25%	LLM	0	0.008	0	-0.012	0.014
NA	25%	Human Validation	0	0.001	0	-0.001	0.001
NA	25%	Debiased	0	0.001	0	-0.002	0.001
NA	50%	LLM	0	0.008	0	-0.012	0.013
NA	50%	Human Validation	0	0.000	0	-0.001	0.001
NA	50%	Debiased	0	0.001	0	-0.001	0.001

Table 19: Normalized Bias Summary of β_3 Statistics by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
NA	5%	LLM	0.148	0.846	0.166	-1.236	1.398
NA	5%	Human Validation	0.001	0.032	0.005	-0.052	0.057
NA	5%	Debiased	0.003	0.033	0.003	-0.051	0.051
NA	10%	LLM	0.150	0.842	0.139	-1.202	1.419
NA	10%	Human Validation	-0.002	0.029	-0.006	-0.051	0.049
NA	10%	Debiased	0.000	0.033	-0.003	-0.056	0.050
NA	25%	LLM	0.154	0.852	0.165	-1.287	1.419
NA	25%	Human Validation	0.006	0.028	0.008	-0.040	0.047
NA	25%	Debiased	-0.001	0.035	-0.001	-0.055	0.057
NA	50%	LLM	0.150	0.845	0.124	-1.286	1.444
NA	50%	Human Validation	0.001	0.032	0.003	-0.053	0.045
NA	50%	Debiased	-0.002	0.029	-0.004	-0.046	0.047

Table 20: Normalized Bias Summary Statistics of β_{14} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
NA	5%	LLM	0.013	1.541	-0.537	-1.885	2.673
NA	5%	Human Validation	-0.002	0.029	-0.001	-0.050	0.044
NA	5%	Debiased	0.004	0.039	0.002	-0.057	0.063
NA	10%	LLM	0.013	1.543	-0.553	-1.929	2.667
NA	10%	Human Validation	-0.006	0.028	-0.006	-0.046	0.043
NA	10%	Debiased	0.002	0.039	0.003	-0.058	0.067
NA	25%	LLM	0.005	1.541	-0.560	-1.934	2.622
NA	25%	Human Validation	0.001	0.032	-0.001	-0.047	0.050
NA	25%	Debiased	-0.001	0.034	0.005	-0.059	0.056
NA	50%	LLM	0.009	1.547	-0.543	-1.968	2.667
NA	50%	Human Validation	0.003	0.032	0.004	-0.042	0.050
NA	50%	Debiased	0.001	0.034	0.001	-0.057	0.057

Table 21: Normalized Bias Summary Statistics of β_{15} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
NA	5%	LLM	0.118	0.463	0.166	-0.633	0.791
NA	5%	Human Validation	-0.003	0.033	-0.004	-0.056	0.050
NA	5%	Debiased	-0.002	0.030	-0.002	-0.046	0.045
NA	10%	LLM	0.108	0.470	0.167	-0.695	0.764
NA	10%	Human Validation	-0.001	0.034	0.000	-0.047	0.056
NA	10%	Debiased	-0.003	0.033	-0.002	-0.060	0.048
NA	25%	LLM	0.121	0.460	0.150	-0.653	0.782
NA	25%	Human Validation	-0.005	0.034	-0.012	-0.055	0.056
NA	25%	Debiased	0.004	0.035	0.007	-0.052	0.054
NA	50%	LLM	0.112	0.450	0.158	-0.649	0.762
NA	50%	Human Validation	-0.006	0.028	-0.003	-0.052	0.035
NA	50%	Debiased	-0.002	0.030	0.000	-0.051	0.042

Table 22: Normalized Bias Summary Statistics of β_{19} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
NA	5%	LLM	0.369	0.615	0.486	-0.534	1.285
NA	5%	Human Validation	0.008	0.034	0.010	-0.050	0.062
NA	5%	Debiased	-0.010	0.034	-0.009	-0.064	0.039
NA	10%	LLM	0.368	0.615	0.466	-0.502	1.344
NA	10%	Human Validation	-0.004	0.029	-0.006	-0.044	0.039
NA	10%	Debiased	-0.004	0.034	-0.003	-0.064	0.048
NA	25%	LLM	0.359	0.607	0.451	-0.493	1.326
NA	25%	Human Validation	-0.001	0.032	-0.001	-0.058	0.055
NA	25%	Debiased	0.000	0.033	-0.002	-0.045	0.054
NA	50%	LLM	0.361	0.619	0.466	-0.554	1.331
NA	50%	Human Validation	0.007	0.027	0.007	-0.034	0.047
NA	50%	Debiased	-0.002	0.035	0.000	-0.069	0.049

Table 23: Normalized Bias Summary Statistics of β_{20} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
NA	5%	LLM	-0.008	0.632	0.023	-1.000	0.888
NA	5%	Human Validation	-0.001	0.034	0.001	-0.059	0.047
NA	5%	Debiased	0.002	0.030	0.001	-0.041	0.048
NA	10%	LLM	-0.002	0.630	0.041	-1.005	0.864
NA	10%	Human Validation	0.001	0.030	0.002	-0.049	0.043
NA	10%	Debiased	0.004	0.033	0.004	-0.048	0.056
NA	25%	LLM	-0.009	0.623	0.029	-0.970	0.838
NA	25%	Human Validation	0.003	0.030	0.004	-0.049	0.047
NA	25%	Debiased	-0.001	0.030	-0.007	-0.040	0.060
NA	50%	LLM	-0.003	0.619	0.049	-0.990	0.835
NA	50%	Human Validation	0.001	0.034	0.000	-0.055	0.054
NA	50%	Debiased	0.000	0.036	0.003	-0.055	0.061

Table 24: Normalized Bias Summary Statistics of β_{Other} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
NA	5%	LLM	-0.093	0.979	0.030	-1.766	1.413
NA	5%	Human Validation	-0.003	0.033	-0.006	-0.051	0.055
NA	5%	Debiased	0.001	0.031	-0.006	-0.046	0.055
NA	10%	LLM	-0.100	0.961	0.021	-1.788	1.394
NA	10%	Human Validation	-0.001	0.030	0.001	-0.049	0.044
NA	10%	Debiased	0.000	0.035	-0.004	-0.049	0.060
NA	25%	LLM	-0.100	0.970	0.016	-1.767	1.439
NA	25%	Human Validation	0.001	0.030	-0.002	-0.045	0.053
NA	25%	Debiased	-0.005	0.036	-0.004	-0.065	0.055
NA	50%	LLM	-0.082	0.957	0.036	-1.721	1.424
NA	50%	Human Validation	-0.001	0.034	0.008	-0.057	0.043
NA	50%	Debiased	0.004	0.037	0.001	-0.050	0.060

Table 25: MSE Summary Statistics of β_3 by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.001	0.001	0.001	0.001	0.002
5%	Human Validation	0.012	0.002	0.013	0.009	0.014
5%	Debiased	0.015	0.015	0.011	0.006	0.031
10%	LLM	0.001	0.001	0.001	0.001	0.002
10%	Human Validation	0.006	0.001	0.006	0.004	0.007
10%	Debiased	0.007	0.003	0.005	0.003	0.013
25%	LLM	0.001	0.001	0.001	0.001	0.003
25%	Human Validation	0.002	0.000	0.002	0.002	0.003
25%	Debiased	0.003	0.001	0.003	0.002	0.006
50%	LLM	0.001	0.001	0.001	0.001	0.002
50%	Human Validation	0.001	0.000	0.001	0.001	0.001
50%	Debiased	0.003	0.001	0.003	0.002	0.005

Table 26: MSE Summary Statistics of β_{14} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.001	0.001	0.001	0.001	0.003
5%	Human Validation	0.014	0.003	0.015	0.010	0.017
5%	Debiased	3.543	23.671	0.099	0.055	2.287
10%	LLM	0.001	0.001	0.001	0.001	0.003
10%	Human Validation	0.007	0.001	0.007	0.005	0.009
10%	Debiased	0.037	0.013	0.036	0.020	0.057
25%	LLM	0.001	0.001	0.001	0.001	0.003
25%	Human Validation	0.003	0.001	0.003	0.002	0.003
25%	Debiased	0.016	0.005	0.015	0.009	0.026
50%	LLM	0.001	0.001	0.001	0.001	0.003
50%	Human Validation	0.001	0.000	0.001	0.001	0.002
50%	Debiased	0.012	0.004	0.012	0.007	0.018

Table 27: MSE Summary Statistics of β_{15} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.001	0.000	0.001	0.000	0.001
5%	Human Validation	0.008	0.002	0.009	0.005	0.011
5%	Debiased	0.018	0.006	0.018	0.009	0.029
10%	LLM	0.001	0.000	0.001	0.000	0.001
10%	Human Validation	0.004	0.001	0.004	0.002	0.005
10%	Debiased	0.008	0.003	0.009	0.004	0.013
25%	LLM	0.001	0.000	0.001	0.000	0.001
25%	Human Validation	0.002	0.000	0.002	0.001	0.002
25%	Debiased	0.004	0.001	0.004	0.002	0.006
50%	LLM	0.001	0.000	0.001	0.000	0.001
50%	Human Validation	0.001	0.000	0.001	0.000	0.001
50%	Debiased	0.003	0.001	0.004	0.002	0.005

Table 28: MSE Summary Statistics of β_{19} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.001	0.000	0.001	0.000	0.001
5%	Human Validation	0.006	0.001	0.006	0.004	0.008
5%	Debiased	0.061	0.369	0.015	0.008	0.038
10%	LLM	0.001	0.000	0.001	0.000	0.001
10%	Human Validation	0.003	0.001	0.003	0.002	0.004
10%	Debiased	0.008	0.003	0.007	0.004	0.012
25%	LLM	0.001	0.000	0.001	0.000	0.001
25%	Human Validation	0.001	0.000	0.001	0.001	0.001
25%	Debiased	0.004	0.001	0.003	0.002	0.006
50%	LLM	0.001	0.000	0.001	0.000	0.001
50%	Human Validation	0.001	0.000	0.001	0.000	0.001
50%	Debiased	0.003	0.001	0.003	0.002	0.005

Table 29: MSE Summary Statistics of β_{20} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.001	0.000	0.001	0.000	0.001
5%	Human Validation	0.007	0.002	0.008	0.005	0.009
5%	Debiased	0.009	0.006	0.009	0.004	0.015
10%	LLM	0.001	0.000	0.001	0.000	0.001
10%	Human Validation	0.004	0.001	0.004	0.002	0.005
10%	Debiased	0.004	0.001	0.004	0.002	0.007
25%	LLM	0.001	0.000	0.001	0.000	0.001
25%	Human Validation	0.001	0.000	0.002	0.001	0.002
25%	Debiased	0.002	0.001	0.002	0.001	0.004
50%	LLM	0.001	0.000	0.001	0.000	0.001
50%	Human Validation	0.001	0.000	0.001	0.000	0.001
50%	Debiased	0.002	0.001	0.002	0.001	0.003

Table 30: MSE Summary Statistics of β_{Other} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.000	0.000	0.000	0.000	0.000
5%	Human Validation	0.002	0.000	0.002	0.001	0.002
5%	Debiased	0.011	0.052	0.002	0.001	0.009
10%	LLM	0.000	0.000	0.000	0.000	0.000
10%	Human Validation	0.001	0.000	0.001	0.001	0.001
10%	Debiased	0.001	0.000	0.001	0.001	0.002
25%	LLM	0.000	0.000	0.000	0.000	0.000
25%	Human Validation	0.000	0.000	0.000	0.000	0.000
25%	Debiased	0.001	0.000	0.001	0.000	0.001
50%	LLM	0.000	0.000	0.000	0.000	0.000
50%	Human Validation	0.000	0.000	0.000	0.000	0.000
50%	Debiased	0.001	0.000	0.001	0.000	0.001

Table 31: MSE Summary Statistics of β_3 by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
NA	5%	LLM	0.001	0.001	0.001	0.001	0.002
NA	5%	Human Validation	0.012	0.002	0.013	0.009	0.014
NA	5%	Debiased	0.015	0.015	0.011	0.006	0.031
NA	10%	LLM	0.001	0.001	0.001	0.001	0.002
NA	10%	Human Validation	0.006	0.001	0.006	0.004	0.007
NA	10%	Debiased	0.007	0.003	0.005	0.003	0.013
NA	25%	LLM	0.001	0.001	0.001	0.001	0.003
NA	25%	Human Validation	0.002	0.000	0.002	0.002	0.003
NA	25%	Debiased	0.003	0.001	0.003	0.002	0.006
NA	50%	LLM	0.001	0.001	0.001	0.001	0.002
NA	50%	Human Validation	0.001	0.000	0.001	0.001	0.001
NA	50%	Debiased	0.003	0.001	0.003	0.002	0.005

Table 32: MSE Summary Statistics of β_{14} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
NA	5%	LLM	0.001	0.001	0.001	0.001	0.003
NA	5%	Human Validation	0.014	0.003	0.015	0.010	0.017
NA	5%	Debiased	3.543	23.671	0.099	0.055	2.287
NA	10%	LLM	0.001	0.001	0.001	0.001	0.003
NA	10%	Human Validation	0.007	0.001	0.007	0.005	0.009
NA	10%	Debiased	0.037	0.013	0.036	0.020	0.057
NA	25%	LLM	0.001	0.001	0.001	0.001	0.003
NA	25%	Human Validation	0.003	0.001	0.003	0.002	0.003
NA	25%	Debiased	0.016	0.005	0.015	0.009	0.026
NA	50%	LLM	0.001	0.001	0.001	0.001	0.003
NA	50%	Human Validation	0.001	0.000	0.001	0.001	0.002
NA	50%	Debiased	0.012	0.004	0.012	0.007	0.018

Table 33: MSE Summary Statistics of β_{15} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
NA	5%	LLM	0.001	0.000	0.001	0.000	0.001
NA	5%	Human Validation	0.008	0.002	0.009	0.005	0.011
NA	5%	Debiased	0.018	0.006	0.018	0.009	0.029
NA	10%	LLM	0.001	0.000	0.001	0.000	0.001
NA	10%	Human Validation	0.004	0.001	0.004	0.002	0.005
NA	10%	Debiased	0.008	0.003	0.009	0.004	0.013
NA	25%	LLM	0.001	0.000	0.001	0.000	0.001
NA	25%	Human Validation	0.002	0.000	0.002	0.001	0.002
NA	25%	Debiased	0.004	0.001	0.004	0.002	0.006
NA	50%	LLM	0.001	0.000	0.001	0.000	0.001
NA	50%	Human Validation	0.001	0.000	0.001	0.000	0.001
NA	50%	Debiased	0.003	0.001	0.004	0.002	0.005

Table 34: MSE Summary Statistics of β_{19} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
NA	5%	LLM	0.001	0.000	0.001	0.000	0.001
NA	5%	Human Validation	0.006	0.001	0.006	0.004	0.008
NA	5%	Debiased	0.061	0.369	0.015	0.008	0.038
NA	10%	LLM	0.001	0.000	0.001	0.000	0.001
NA	10%	Human Validation	0.003	0.001	0.003	0.002	0.004
NA	10%	Debiased	0.008	0.003	0.007	0.004	0.012
NA	25%	LLM	0.001	0.000	0.001	0.000	0.001
NA	25%	Human Validation	0.001	0.000	0.001	0.001	0.001
NA	25%	Debiased	0.004	0.001	0.003	0.002	0.006
NA	50%	LLM	0.001	0.000	0.001	0.000	0.001
NA	50%	Human Validation	0.001	0.000	0.001	0.000	0.001
NA	50%	Debiased	0.003	0.001	0.003	0.002	0.005

Table 35: MSE Summary Statistics of β_{20} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
NA	5%	LLM	0.001	0.000	0.001	0.000	0.001
NA	5%	Human Validation	0.007	0.002	0.008	0.005	0.009
NA	5%	Debiased	0.009	0.006	0.009	0.004	0.015
NA	10%	LLM	0.001	0.000	0.001	0.000	0.001
NA	10%	Human Validation	0.004	0.001	0.004	0.002	0.005
NA	10%	Debiased	0.004	0.001	0.004	0.002	0.007
NA	25%	LLM	0.001	0.000	0.001	0.000	0.001
NA	25%	Human Validation	0.001	0.000	0.002	0.001	0.002
NA	25%	Debiased	0.002	0.001	0.002	0.001	0.004
NA	50%	LLM	0.001	0.000	0.001	0.000	0.001
NA	50%	Human Validation	0.001	0.000	0.001	0.000	0.001
NA	50%	Debiased	0.002	0.001	0.002	0.001	0.003

Table 36: MSE Summary Statistics of β_{Other} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
NA	5%	LLM	0.000	0.000	0.000	0.000	0.000
NA	5%	Human Validation	0.002	0.000	0.002	0.001	0.002
NA	5%	Debiased	0.011	0.052	0.002	0.001	0.009
NA	10%	LLM	0.000	0.000	0.000	0.000	0.000
NA	10%	Human Validation	0.001	0.000	0.001	0.001	0.001
NA	10%	Debiased	0.001	0.000	0.001	0.001	0.002
NA	25%	LLM	0.000	0.000	0.000	0.000	0.000
NA	25%	Human Validation	0.000	0.000	0.000	0.000	0.000
NA	25%	Debiased	0.001	0.000	0.001	0.000	0.001
NA	50%	LLM	0.000	0.000	0.000	0.000	0.000
NA	50%	Human Validation	0.000	0.000	0.000	0.000	0.000
NA	50%	Debiased	0.001	0.000	0.001	0.000	0.001

Table 37: Coverage Summary Statistics of β_3 by Proportion of Validation Samples. Proxy on the RHS:
 $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.867	0.096	0.911	0.689	0.952
5%	Human Validation	0.919	0.008	0.919	0.906	0.931
5%	Debiased	0.920	0.011	0.921	0.904	0.939
10%	LLM	0.866	0.097	0.913	0.679	0.947
10%	Human Validation	0.935	0.008	0.935	0.923	0.947
10%	Debiased	0.940	0.007	0.940	0.928	0.952
25%	LLM	0.865	0.097	0.909	0.692	0.948
25%	Human Validation	0.944	0.008	0.944	0.931	0.954
25%	Debiased	0.945	0.007	0.946	0.934	0.955
50%	LLM	0.865	0.099	0.913	0.653	0.947
50%	Human Validation	0.949	0.007	0.948	0.939	0.959
50%	Debiased	0.948	0.007	0.949	0.937	0.957

Table 38: Coverage Summary Statistics of β_{14} by Proportion of Validation Samples. Proxy on the RHS:
 $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.685	0.225	0.764	0.252	0.932
5%	Human Validation	0.917	0.009	0.917	0.901	0.930
5%	Debiased	0.938	0.010	0.939	0.920	0.953
10%	LLM	0.689	0.226	0.770	0.242	0.937
10%	Human Validation	0.933	0.008	0.934	0.918	0.945
10%	Debiased	0.940	0.008	0.942	0.926	0.951
25%	LLM	0.686	0.225	0.759	0.237	0.938
25%	Human Validation	0.943	0.008	0.944	0.929	0.955
25%	Debiased	0.946	0.007	0.946	0.937	0.957
50%	LLM	0.690	0.225	0.766	0.255	0.935
50%	Human Validation	0.947	0.007	0.948	0.936	0.959
50%	Debiased	0.948	0.007	0.949	0.937	0.958

Table 39: Coverage Summary Statistics of β_{15} by Proportion of Validation Samples. Proxy on the RHS:
 $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.925	0.032	0.936	0.879	0.957
5%	Human Validation	0.932	0.009	0.931	0.919	0.946
5%	Debiased	0.932	0.010	0.933	0.919	0.950
10%	LLM	0.926	0.030	0.938	0.876	0.953
10%	Human Validation	0.940	0.008	0.941	0.927	0.951
10%	Debiased	0.941	0.008	0.941	0.928	0.955
25%	LLM	0.925	0.033	0.938	0.868	0.953
25%	Human Validation	0.946	0.007	0.946	0.934	0.958
25%	Debiased	0.947	0.007	0.947	0.934	0.957
50%	LLM	0.926	0.031	0.933	0.883	0.956
50%	Human Validation	0.947	0.006	0.947	0.938	0.958
50%	Debiased	0.947	0.007	0.947	0.934	0.957

Table 40: Coverage Summary Statistics of β_{19} by Proportion of Validation Samples. Proxy on the RHS:
 $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.891	0.065	0.915	0.740	0.954
5%	Human Validation	0.937	0.009	0.938	0.926	0.953
5%	Debiased	0.944	0.009	0.944	0.931	0.956
10%	LLM	0.890	0.066	0.912	0.750	0.949
10%	Human Validation	0.943	0.008	0.943	0.930	0.954
10%	Debiased	0.947	0.008	0.947	0.934	0.957
25%	LLM	0.894	0.064	0.915	0.737	0.954
25%	Human Validation	0.947	0.007	0.947	0.938	0.958
25%	Debiased	0.947	0.008	0.948	0.935	0.959
50%	LLM	0.889	0.066	0.913	0.735	0.947
50%	Human Validation	0.951	0.007	0.950	0.940	0.961
50%	Debiased	0.947	0.007	0.948	0.935	0.957

Table 41: Coverage Summary Statistics of β_{20} by Proportion of Validation Samples. Proxy on the RHS:
 $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.901	0.062	0.923	0.784	0.956
5%	Human Validation	0.938	0.007	0.938	0.926	0.950
5%	Debiased	0.938	0.008	0.938	0.926	0.951
10%	LLM	0.902	0.056	0.920	0.785	0.953
10%	Human Validation	0.943	0.007	0.942	0.932	0.956
10%	Debiased	0.945	0.007	0.944	0.934	0.956
25%	LLM	0.903	0.058	0.925	0.783	0.952
25%	Human Validation	0.948	0.007	0.948	0.937	0.959
25%	Debiased	0.947	0.007	0.947	0.937	0.959
50%	LLM	0.904	0.055	0.923	0.796	0.954
50%	Human Validation	0.950	0.007	0.951	0.938	0.960
50%	Debiased	0.949	0.007	0.948	0.939	0.961

Table 42: Coverage Summary Statistics of β_{Other} by Proportion of Validation Samples. Proxy on the RHS:
 $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.836	0.133	0.886	0.570	0.950
5%	Human Validation	0.949	0.007	0.949	0.937	0.961
5%	Debiased	0.954	0.007	0.954	0.943	0.966
10%	LLM	0.841	0.133	0.891	0.550	0.955
10%	Human Validation	0.950	0.007	0.950	0.939	0.960
10%	Debiased	0.951	0.008	0.952	0.938	0.962
25%	LLM	0.839	0.133	0.887	0.581	0.949
25%	Human Validation	0.949	0.007	0.950	0.938	0.958
25%	Debiased	0.948	0.007	0.950	0.935	0.959
50%	LLM	0.840	0.129	0.891	0.561	0.951
50%	Human Validation	0.951	0.007	0.950	0.940	0.962
50%	Debiased	0.949	0.007	0.949	0.939	0.960

Table 43: Coverage Summary Statistics of β_3 by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
NA	5%	LLM	0.867	0.096	0.911	0.689	0.952
NA	5%	Human Validation	0.919	0.008	0.919	0.906	0.931
NA	5%	Debiased	0.920	0.011	0.921	0.904	0.939
NA	10%	LLM	0.866	0.097	0.913	0.679	0.947
NA	10%	Human Validation	0.935	0.008	0.935	0.923	0.947
NA	10%	Debiased	0.940	0.007	0.940	0.928	0.952
NA	25%	LLM	0.865	0.097	0.909	0.692	0.948
NA	25%	Human Validation	0.944	0.008	0.944	0.931	0.954
NA	25%	Debiased	0.945	0.007	0.946	0.934	0.955
NA	50%	LLM	0.865	0.099	0.913	0.653	0.947
NA	50%	Human Validation	0.949	0.007	0.948	0.939	0.959
NA	50%	Debiased	0.948	0.007	0.949	0.937	0.957

Table 44: Coverage Summary Statistics of β_{14} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
NA	5%	LLM	0.685	0.225	0.764	0.252	0.932
NA	5%	Human Validation	0.917	0.009	0.917	0.901	0.930
NA	5%	Debiased	0.938	0.010	0.939	0.920	0.953
NA	10%	LLM	0.689	0.226	0.770	0.242	0.937
NA	10%	Human Validation	0.933	0.008	0.934	0.918	0.945
NA	10%	Debiased	0.940	0.008	0.942	0.926	0.951
NA	25%	LLM	0.686	0.225	0.759	0.237	0.938
NA	25%	Human Validation	0.943	0.008	0.944	0.929	0.955
NA	25%	Debiased	0.946	0.007	0.946	0.937	0.957
NA	50%	LLM	0.690	0.225	0.766	0.255	0.935
NA	50%	Human Validation	0.947	0.007	0.948	0.936	0.959
NA	50%	Debiased	0.948	0.007	0.949	0.937	0.958

Table 45: Coverage Summary Statistics of β_{15} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
NA	5%	LLM	0.925	0.032	0.936	0.879	0.957
NA	5%	Human Validation	0.932	0.009	0.931	0.919	0.946
NA	5%	Debiased	0.932	0.010	0.933	0.919	0.950
NA	10%	LLM	0.926	0.030	0.938	0.876	0.953
NA	10%	Human Validation	0.940	0.008	0.941	0.927	0.951
NA	10%	Debiased	0.941	0.008	0.941	0.928	0.955
NA	25%	LLM	0.925	0.033	0.938	0.868	0.953
NA	25%	Human Validation	0.946	0.007	0.946	0.934	0.958
NA	25%	Debiased	0.947	0.007	0.947	0.934	0.957
NA	50%	LLM	0.926	0.031	0.933	0.883	0.956
NA	50%	Human Validation	0.947	0.006	0.947	0.938	0.958
NA	50%	Debiased	0.947	0.007	0.947	0.934	0.957

Table 46: Coverage Summary Statistics of β_{19} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
NA	5%	LLM	0.891	0.065	0.915	0.740	0.954
NA	5%	Human Validation	0.937	0.009	0.938	0.926	0.953
NA	5%	Debiased	0.944	0.009	0.944	0.931	0.956
NA	10%	LLM	0.890	0.066	0.912	0.750	0.949
NA	10%	Human Validation	0.943	0.008	0.943	0.930	0.954
NA	10%	Debiased	0.947	0.008	0.947	0.934	0.957
NA	25%	LLM	0.894	0.064	0.915	0.737	0.954
NA	25%	Human Validation	0.947	0.007	0.947	0.938	0.958
NA	25%	Debiased	0.947	0.008	0.948	0.935	0.959
NA	50%	LLM	0.889	0.066	0.913	0.735	0.947
NA	50%	Human Validation	0.951	0.007	0.950	0.940	0.961
NA	50%	Debiased	0.947	0.007	0.948	0.935	0.957

Table 47: Coverage Summary Statistics of β_{20} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
NA	5%	LLM	0.901	0.062	0.923	0.784	0.956
NA	5%	Human Validation	0.938	0.007	0.938	0.926	0.950
NA	5%	Debiased	0.938	0.008	0.938	0.926	0.951
NA	10%	LLM	0.902	0.056	0.920	0.785	0.953
NA	10%	Human Validation	0.943	0.007	0.942	0.932	0.956
NA	10%	Debiased	0.945	0.007	0.944	0.934	0.956
NA	25%	LLM	0.903	0.058	0.925	0.783	0.952
NA	25%	Human Validation	0.948	0.007	0.948	0.937	0.959
NA	25%	Debiased	0.947	0.007	0.947	0.937	0.959
NA	50%	LLM	0.904	0.055	0.923	0.796	0.954
NA	50%	Human Validation	0.950	0.007	0.951	0.938	0.960
NA	50%	Debiased	0.949	0.007	0.948	0.939	0.961

Table 48: Coverage Summary Statistics of β_{Other} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
NA	5%	LLM	0.836	0.133	0.886	0.570	0.950
NA	5%	Human Validation	0.949	0.007	0.949	0.937	0.961
NA	5%	Debiased	0.954	0.007	0.954	0.943	0.966
NA	10%	LLM	0.841	0.133	0.891	0.550	0.955
NA	10%	Human Validation	0.950	0.007	0.950	0.939	0.960
NA	10%	Debiased	0.951	0.008	0.952	0.938	0.962
NA	25%	LLM	0.839	0.133	0.887	0.581	0.949
NA	25%	Human Validation	0.949	0.007	0.950	0.938	0.958
NA	25%	Debiased	0.948	0.007	0.950	0.935	0.959
NA	50%	LLM	0.840	0.129	0.891	0.561	0.951
NA	50%	Human Validation	0.951	0.007	0.950	0.940	0.962
NA	50%	Debiased	0.949	0.007	0.949	0.939	0.960